Single-run SHAP reliability under finite computation

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Abstract

A stochastic SHAP attribution is a numerical estimate, even when displayed as a list of reasons for a prediction. We assess when a single run preserves a fixed empirical-game attribution under finite CPU budgets. Across three public tabular datasets, three fixed classifiers and two estimators, we compare 97,200 explanations with 540 exactly enumerated references. Our declared adequacy event requires normalized L1 error at most .20 and recovery of at least two of three signed salient features. Lowest-budget Kernel success ranges from .7794 to 1 across fitted cells; every highest-budget condition has observed success 1, with a conservative boundary interval [.8247,1] over 60 input clusters. This does not certify population or individual reliability. Requiring all three signed features reduces highest-budget success as low as .9289, and a lower-budget input has 0/30 successes despite high panel-average performance. Repeatability and reference accuracy are separately measured. The main experiment uses 7.493 CPU hours; interrupted elapsed clocks are retained and cannot establish deployment latency. The results support conditional numerical adequacy for the tested games, budgets and readings, with explicit failure cases and limits on transfer to consequential decisions.

1 Introduction

A local feature attribution can be read as a list of reasons for a classifier's prediction: which features contribute most, in which direction, and by how much. When the attribution is computed by a stochastic approximation, the displayed list is also a numerical estimate. Repeating the explainer while holding the prediction problem fixed can change that estimate. For credit or health applications, understanding this source of variation is a prerequisite to interpreting the display; it is not by itself evidence that the prediction or its explanation is suitable for a consequential decision.

The SHAP framework defines additive attributions and supplies algorithms with different computational properties [Lundberg and Lee, 2017]. Its original evaluation already relates sampled-estimator error to evaluation count, and subsequent work studies KernelSHAP uncertainty and convergence [Covert and Lee, 2021]. The practical question here is conditional: for a specified trained tabular model, input, empirical background and computation budget, how often does *one* approximate attribution preserve both the reference vector's magnitude and its salient signed features? This question requires more than checking whether repeated runs look similar. Agreement among approximations, error relative to a common target, and the adequacy of an explanation for a particular use are distinct properties.

We organize the study around three questions. First, how does the frequency of meeting an explicit one-run numerical criterion change with actual prediction work and CPU time? Second, how closely does agreement between independent runs track accuracy against the same-game reference? Third, do reference-defined weak signals, small importance gaps and numerical ties identify conditions in which the assessment changes? The adequacy thresholds are declared study conventions, with component and sensitivity analyses; they are not accepted lending or clinical standards.

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The completed study combines three public tabular tasks with logistic regression, random forests and histogram gradient boosting. For each fitted model, a fixed held-out panel and empirical background isolate explainer randomness. Grouped Kernel SHAP and paired permutation sampling are compared with exact enumeration of the same empirical marginal game. Repeated seeds quantify within-input variability, while inference over inputs preserves their clustering. Requested computation settings are accompanied by actual model-input rows and measured time, because background expansion, cached coalitions and exhaustive evaluation can make nominal settings misleading.

The study's contribution is a reproducible assessment of conditional one-run numerical adequacy, with explicit targets, costs, uncertainty and failure conditions. It does not introduce a new Shapley estimator or claim to discover stochastic attribution error. The measured panel shows budget- and criterion-dependent adequacy: lowest-budget Kernel success spans $.7794$ to 1, whereas all highest-budget conditions meet the broad event in every observed run. Stricter signed-feature recovery and individual-input failures qualify that result. Conclusions remain conditional on the fitted predictors, tasks, backgrounds, implementations and tested computation range.

2 Related work

Finite-budget Shapley estimation. The SHAP framework connects additive feature attribution to Shapley values and provides model-agnostic sampling and weighted-regression approximations [Lundberg and Lee, 2017]. Its axiomatic properties characterize a specified target, whereas a finite stochastic estimate also has numerical error. Its original computational evaluation already compares estimates with reference values as model-evaluation count grows. Covert and Lee [2021] analyze regression-based Shapley estimators, derive an unbiased alternative, and develop paired sampling, uncertainty estimation and convergence detection. Musco and Witter [2025] propose leverage-score sampling with a regression-objective guarantee and a derived coefficient-error bound under their assumptions. Such guarantees do not directly establish signed top-feature recovery for every prediction. Our comparison covers Kernel SHAP and paired permutation sampling; it does not rank all newer estimators. Budget dependence, estimator variability and uncertainty estimation are established concerns. The present study targets the probability that one run meets explicit magnitude and salient-feature criteria at measured cost.

Different sources of disagreement. Chen et al. [2020] distinguish observational and marginal attribution games and show why their suitability depends on the application. Krishna et al. [2024] study disagreement among distinct explanation methods, using practitioner-motivated feature, rank and sign comparisons. Yuan et al. [2023] vary DeepExplainer background samples and examine local magnitude variation and global ranking stability. These comparisons change the method or background; our primary repeated-run question holds both fixed. A background sensitivity is a separate target comparison. Similarly, predictive multiplicity concerns competing fitted models [Marx et al., 2020], rather than explanation randomness for a frozen prediction. Several model families broaden the tested settings without providing uncertainty over model training.

Metrics and benchmark scope. Nogueira et al. [2018] formalize feature-subset stability and its uncertainty, emphasizing the sampling unit and metric properties. Webber et al. [2010] give a top-weighted rank comparison with explicit finite-prefix and tie conventions. Membership, ranks, signs and value precision capture different properties; none alone supplies a universal explanation-adequacy cutoff. In particular, overlap divided by k differs from Jaccard overlap, and its chance behavior varies with feature count. We retain value error and signed selection as separate components rather than comparing a single stability score indiscriminately across dimensions. Dataset identity also matters: the ACS tasks of Ding et al. [2021] explicitly specify prediction targets and population filters. We fix state, year, task and weighting policy and limit conclusions to that benchmark. The contribution sought here is a reproducible conditional assessment of one-run numerical adequacy; no priority claim about discovering SHAP variability is made.

3 Attribution target and reliability

Let f be a trained classifier's probability for the selected class, x a held-out input, $N = \{1, \ldots, d\}$ the original-feature players, and $B = (z^{(1)}, \ldots, z^{(m)})$ a fixed training-data background. We use the

empirical marginal game

$$v_{x,B}(S) = \frac{1}{m}\sum_{b=1}^{m} f\left(x_S, z^{(b)}_{N\setminus S}\right), \qquad S \subseteq N. \tag{1}$$

Hybrid records pass through the fitted preprocessing pipeline; an entire categorical feature remains one player even if encoded in several columns. The game's Shapley vector ϕ allocates $f(x) - v_{x,B}(\emptyset)$ in probability units [Lundberg and Lee, 2017]. Appendix C gives the full definition and diagnostics. Exactness means this finite empirical game, up to numerical precision. It does not make the background representative or resolve the application-dependent choice between marginal and observational games [Chen et al., 2020].

Write $\widehat{\phi}_{q,r}$ for an approximation with computation setting q and seed r. Holding the model, input, background, preprocessing, output and feature groups fixed isolates explainer randomness. *Repeatability* compares independent runs; *accuracy* compares a run with a validated same-game reference; *adequacy* asks whether errors satisfy a specified numerical use criterion. Two runs can agree while sharing error. The efficiency residual checks total allocation, not coordinate accuracy. Neither exactness nor repeatability establishes causal validity, fairness or deployment safety. Cost includes executed model-input rows after background expansion and measured time, alongside requested work.

4 Assessment criteria

For a reference $\phi^\star$ for the same game, let $T_k(a)$ select the k largest absolute coordinates, with original-feature index breaking exact ties. Signed overlap $O_k^{\pm}$ is the number of shared selected features with matching contribution signs divided by k; unsigned overlap ignores signs. We combine value error and signed selection in a one-run event:

$$\begin{aligned} E_{\mathrm{rel}}(\widehat{\phi}, \phi^\star) &= \frac{\|\widehat{\phi} - \phi^\star\|_1}{\max(\|\phi^\star\|_1, \delta)}, \\ A_{\tau,\eta}(\widehat{\phi}, \phi^\star) &= \mathbf{1}\{E_{\mathrm{rel}} \le \tau \text{ and } O_k^{\pm} \ge \eta\}. \end{aligned} \tag{2}$$

The resolution floor δ, cutoff τ, selection requirement η and k are predeclared study choices, not validated clinical or lending tolerances. Magnitude error, unsigned/signed selection, top-1 identity and rank diagnostics remain separate components. Absolute error, reference norms and floor sensitivity expose weak-signal effects; deterministic point scores retain tied and near-zero cases with explicit ambiguity diagnostics. Appendix C.2 gives full formulas and inference details.

For n inputs with R independent seeds each, the observed panel adequacy is the average of per-input success fractions, $\widehat{p} = n^{-1}\sum_i R^{-1}\sum_r A_{\tau,\eta}(\widehat{\phi}_{i,r}, \phi_i^\star)$. All distinct run pairs are reduced to one repeatability summary per input; shared pairs do not enlarge the independent sample. Pairwise L1 disagreement uses the same reference-norm floor as accuracy; normalization by the runs' own norms is separately labeled. Input-cluster resampling preserves all repeats and matched comparisons. These estimates condition on the fitted model, split, background and implementation, with per-input distributions alongside panel means.

Boundary all-zero/all-one bounded panels retain raw bootstrap endpoints but display positive-width Hoeffding intervals. A separate one-sided bounded-mean guard governs population reliability claims under independent input-level observations. With 60 inputs, even no observed failures cannot certify population success of at least .95 using that guard. Thirty seeds likewise do not certify each input. Unresolved references or incomplete runs cannot silently leave denominators; measured negative outcomes and missing evidence remain distinct.

5 Experimental design

The pre-execution protocol and `analysis/configs/main.json` freeze the design below. Section 6 reports achieved coverage, reference checks, predictive diagnostics and measured costs; raw checkpoints and the evidence ledger preserve their provenance.

Data and fitted models. We use Statlog German Credit, ACSIncome from California's 2018 one-year ACS PUMS, and OpenML Diabetes (data ID 37, version 1), with 20, 10 and 8 original-feature players, respectively [Hofmann, 1994, U.S. Census Bureau, 2018, Ding et al., 2021, OpenML, 2014]. German Credit explains the probability of bad credit (original class 2); ACSIncome explains income above $50,000; Diabetes explains the positive label. The ACS task uses folktables 0.0.12 and its working-adult filter: age greater than 16, income greater than $100, positive weekly work hours and person weight at least one. Survey weights do not weight the benchmark. This is a person-row prediction experiment, not a population-representative ACS analysis. Diabetes retains the loader's declared treatment of zero plasma glucose, blood pressure, skin thickness, insulin and body mass index as missing; pregnancy counts of zero remain valid.

Each dataset uses a fixed stratified 80/20 training/test split; ACS training is capped at 20,000 by stratified subsampling. Preprocessing is fitted only on training rows: median imputation and standardization for numeric fields, most-frequent imputation for categorical fields, and one-hot encoding for logistic regression or ordinal encoding for tree models. The model families are logistic regression, a 100-tree random forest and 100-iteration histogram gradient boosting. They provide contrasting CPU-feasible predictors; this is neither a model-selection benchmark nor a survey of all tabular architectures. Training seed 20261103 is fixed. Test accuracy, ROC AUC and log loss are retained for every fitted model, separately from attribution quality.

Inputs, targets and estimators. A uniform draw of 32 training rows defines each fixed background. A uniform held-out panel is shared across model types within a dataset; the first two selected inputs are reserved for pilots and excluded from the 60 main inputs. No selection uses explanation outcomes. Each input/estimator/budget has 30 independently seeded runs, with model, background, feature groups and probability output fixed. Seeds derive from base 314159 and dataset/model/input/estimator/budget/run identifiers. The design compares grouped Kernel SHAP (SHAP 0.50.0, identity link, `l1_reg=0`) with antithetic permutation sampling, which averages a random feature order and its reverse. It does not evaluate every available Shapley estimator.

Requested coalition-work settings are 128/512/2048 for German Credit, 64/256/512 for ACSIncome, and 32/64/128 for Diabetes. These dimension-specific settings are intended to span finite approximation budgets while avoiding needless exhaustion of smaller games. Permutation runs use an even number of paths and cache repeated coalitions within a run. Kernel SHAP records the effective varying-player count, sampled coalitions and design-rank diagnostics. Nominal settings need not execute equal model-row work: background expansion, cached coalitions, endpoints, integer paths and exhaustive cells all affect cost. Comparisons therefore require actual prediction rows and measured CPU/elapsed time alongside the nominal setting. Reference and pilot costs belong in total study accounting, separately from one-run cost.

The accepted reference strategy enumerates every coalition of every selected empirical game, using the same model-output adapter and grouped features as the approximations. Batched prediction bounds memory. Exact enumeration is a target-accuracy control; an efficiency check alone cannot validate its coordinates. Appendix D reports completed numerical checks and an unsuccessful exploratory control, alongside the completed main reference audit. The completed matrix contains nine fitted dataset/model cells, 540 reference games and 97,200 stochastic calls. The saved-panel audit verifies these achieved denominators. Missing or failed runs cannot be scored as completed negative results, and neither pilot-only cases nor partial-debug aggregates enter the main evidence.

Operational criteria and uncertainty. The primary scenario in Eq. (2) fixes $k = 3$, $\delta = .01$ probability units, $\tau = .20$ and $\eta = 2/3$. It asks for bounded vector error and agreement on at least two signed salient features. It is an explicitly selected numerical tolerance, without a clinical or lending interpretation. Sensitivities include error cutoffs $.10/.20/.40$, complete signed top-three recovery, a 10^{-6} probability-unit tie/sign tolerance, normalization floors $.001/.01/.05$ and weak-signal cases. Primary exact ties use original-feature index order. Reference-defined strata distinguish $\|\phi^\star\|_1 < .01$ and a normalized largest-versus-second-largest magnitude gap at most $.02$; such strata describe conditions rather than new independent experiments.

Pointwise 95% descriptive intervals use 2,000 instance-cluster percentile bootstrap resamples (seed 8127) of per-input run means. Paired contrasts preserve input identity. The inference is conditional on the fitted model, split and background; it does not average over training or background resampling. Intervals are not simultaneous across the grid. Boundary-rate handling and the limits of finite-panel

certification are stated in Section 4. A separate post-main background-size/resampling arm changes the target and remains outside the fixed-background repetition estimand (Appendix E).

6 Measured one-run reliability

Completed coverage and reference checks. All nine fitted dataset/model cells completed the frozen panel: 540 exact empirical-game references and 97,200 stochastic calls, with 60 inputs and 30 independent seeds per estimator/budget condition. The saved-vector audit verified the full grid, unique seeds, frozen source identity and recorded primary metrics. No failed attempts, warnings, rank-deficient Kernel designs or exhaustive primary calls were recorded. Maximum reference and stochastic efficiency residuals were 6.66×10^{-16} and 8.88×10^{-16} probability units. These residuals supplement the independent implementation checks in Appendix D; they are not standalone coordinate-accuracy proofs. Held-out ROC AUC ranges from .823 to .852 on German Credit, .885 to .896 on ACSIncome and .879 to .894 on Diabetes. These diagnostics describe the fitted predictors, not the correctness of their rationales.

Figure 1 shows adequacy against actual prediction work; Figure 2 contrasts stricter recovery and input-level failures. The matched-scale reference-error/run-disagreement display (Figure 6), CPU curves and complete condition/component/instance displays and tables are in Appendix F.

Finite-budget failures and improvements. For the declared event, lowest-budget Kernel success varies from .7794 to 1 across the nine fitted cells; permutation success varies from .9261 to 1. Diabetes histogram boosting with Kernel at budget 32 is the lowest observed condition: .7794 with pointwise 95% interval $[.7217, .8345]$. ACS forest with Kernel at budget 64 achieves .8544 $[.7767, .9233]$, and German histogram boosting at budget 128 achieves .9494 $[.9133, .9800]$. The differences across these fixed predictors show why a single pooled success rate would conceal useful conditions. Logistic probability explanations also remain stochastic: for example, ACS logistic Kernel at budget 64 achieves .9589 $[.9200, .9883]$.

Increasing work improves the difficult low-budget conditions. For Diabetes histogram boosting with Kernel, budget 32 minus budget 128 has paired-input adequacy difference $-.2206$, interval $[-.2783, -.1655]$. All eighteen highest-budget estimator/model conditions have observed primary success 1: each has 1,800 calls on 60 input clusters, rather than 1,800 independent inputs. Their displayed boundary-adjusted intervals are $[.8247, 1]$. This wider boundary interval reflects the predeclared conservative treatment, not a decline in the observed success or an estimate of zero sampling uncertainty. None clears the separate population-success guard at .95.

The numerical reading changes the answer. The primary criterion permits one of the three signed salient features to differ. Requiring all three, at the same .20 magnitude tolerance, reduces highest-budget observed success to between .9289 and 1. The lowest such strict result is Diabetes histogram boosting with permutation, .9289 $[.8967, .9589]$. Thus high-budget success for the two-of-three reading does not mean exact feature-list recovery. Component, error-cutoff, strict-selection and tie-aware outputs accompany the main event; magnitude-only adequacy must not be described as joint adequacy.

Panel success also conceals individual difficulty. ACS forest Kernel at budget 64 has one held-out input (panel index 30, with its original source row retained in the results) with 0/30 adequate runs. Its per-input Wilson interval is $[0, .1135]$, despite the panel estimate .8544. This case was found retrospectively; it is evidence for displaying input-level variation, not a prospectively validated rule for identifying unreliable inputs. Conversely, thirty successful seeds on another input would not establish absence of rare failures.

Repeatability is a separate measured property. Pairwise disagreement and reference error use the same reference-norm floor, with all run pairs reduced to one input summary. Their agreement events still ask different questions. ACS forest Kernel at budget 64 has one-run reference adequacy .8544, whereas pairwise joint agreement is .7207 $[.6239, .8085]$. This example does not show that reassuring repeatability concealed error; here pairwise agreement is lower. It shows that the two probabilities are not interchangeable. Full vector and signed-selection summaries preserve both assessments, without treating shared run pairs as independent observations. Averaging thirty runs is separately available with its thirty-run cost; its accuracy cannot be credited to one run.

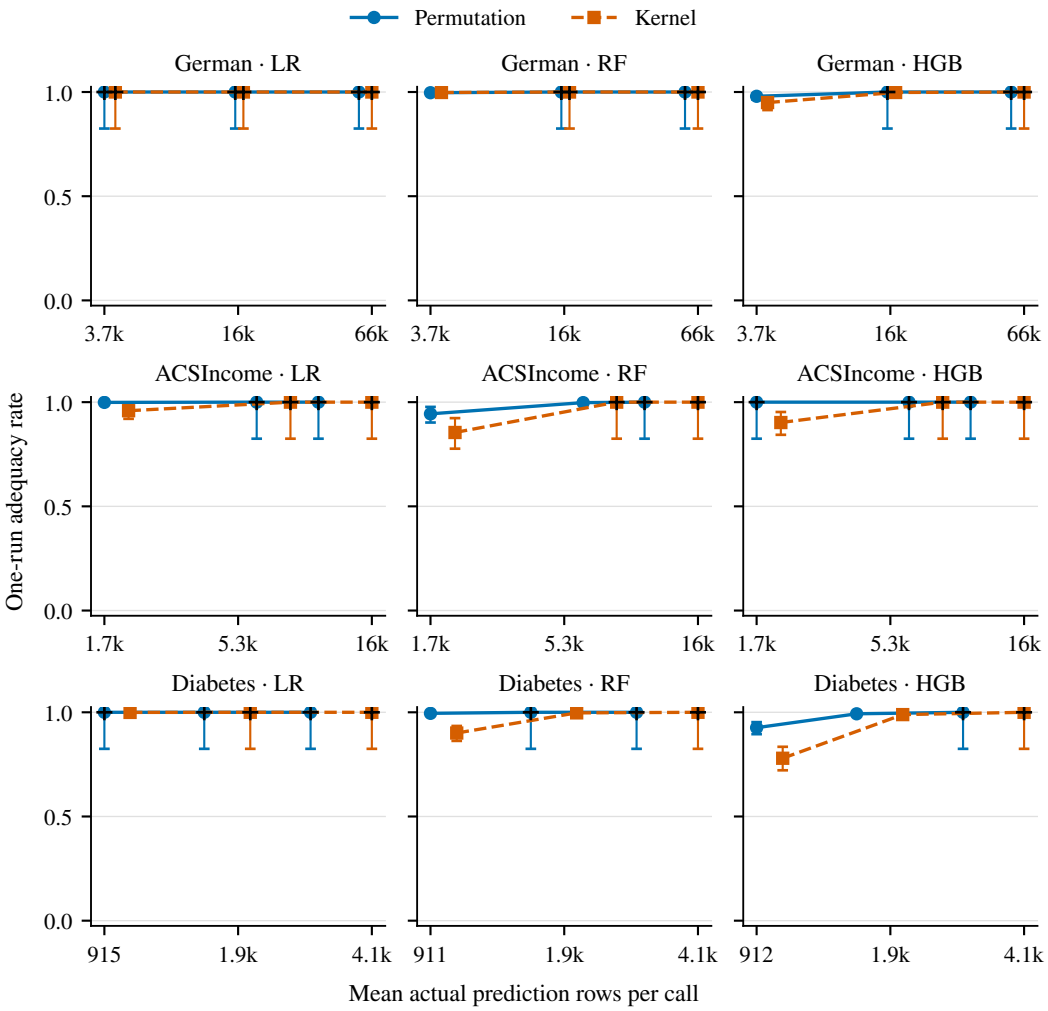


Figure 1: One-run adequacy: normalized L1 error $\leq$0.20 and signed top-3 overlap $\geq$2/3 against the exact finite-game reference. L1 normalization is max(reference L1,0.01 probability units). For each fixed fitted model, 60 held-out inputs and 30 independent runs per input, estimator and requested budget; background B=32. The target is positive-class probability for the fixed empirical interventional game over grouped raw features. Estimates average equally over inputs and runs. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers identify observed all-zero/all-one panels whose displayed bars instead use the supplied two-sided 95% bounded-instance Hoeffding interval. These are observed rates and descriptive intervals, not population 95% adequacy certification. Horizontal positions are mean measured prediction rows; nominal budget equality does not imply equal actual work.

Which failure conditions were observed? Budget, fitted predictor and the requested numerical reading distinguish observed success from failure in this panel. The prespecified near-top-one-gap stratum contains only 2–21 inputs per fitted cell. For example, lowest-budget ACS forest Kernel success within its four near-gap inputs is .6500 [.4417, .8917]; other cells do not show a uniform gap effect. These descriptive strata neither establish a mechanism nor validate a general risk classifier. No reference-defined low-signal input, numerical top-one/top-three boundary tie or selected near-zero sign was observed. The minimum reference L1 norm is .07588, above all tested floors .001/.01/.05; those floor choices therefore give identical results. Low-signal behavior remains untested, rather than demonstrated harmless. Stable index ordering defines exact point scores, and the reported ambiguity counters disclose where numerical tolerance could have affected their interpretation.

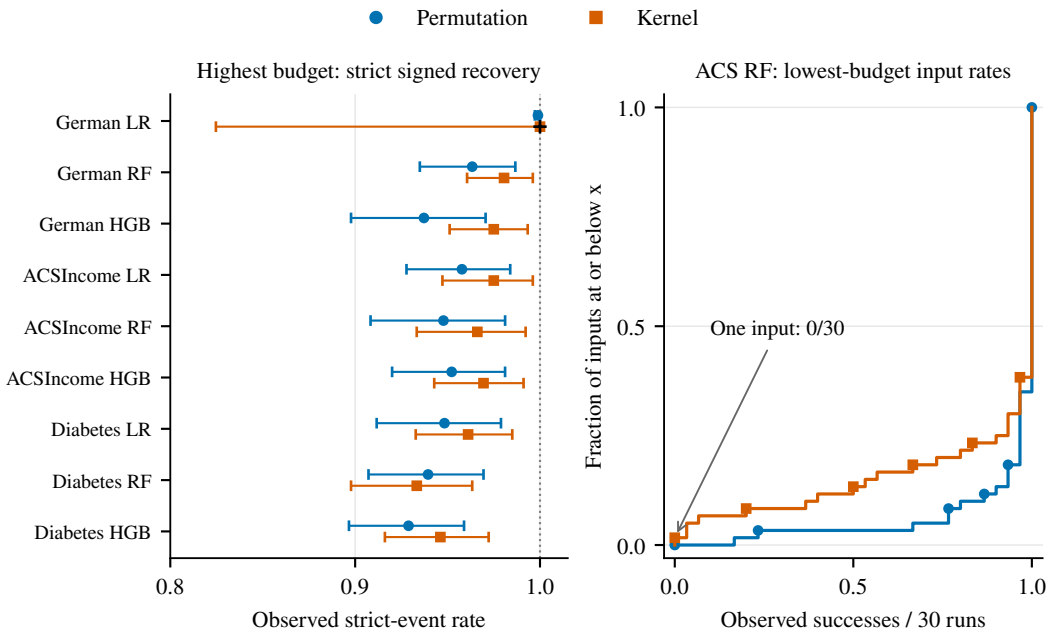


Figure 2: Criterion and input dependence. Left: the strict event at the highest requested budgets (German 2048, ACS 512, Diabetes 128) requires normalized L1 <=0.20 and all three signed top-3 features matching the exact reference. Broad two-of-three primary-event rates are observed 1 for all these conditions, with supplied boundary interval [0.824670,1]; they do not imply exact feature recovery or population >=0.95 certification. Strict-event bars are supplied pointwise 95% instance-cluster percentile intervals; black plus marks denote the supplied two-sided Hoeffding replacement at an observed all-success boundary. Each condition has 60 inputs and 30 independent runs/input. Right: ECDF of observed primary-event input success fractions for ACS RF at requested budget 64; means can hide poor individual observations. Kernel input 30 (original source row 296124) has 0/30 successes, with supplied Wilson interval [0.000,0.114]. The ECDF has no confidence band and is not the distribution of known latent input probabilities. All panels condition on fixed fitted models, split and B32 empirical interventional probability game over raw-feature groups, with L1 normalization max(exact-reference L1,0.01). No individual certificate or equal-cost superiority follows.

Changing the background changes the target. A separate arm retains twelve main inputs per fitted ACS/Diabetes model and compares B8/B128 training-row backgrounds at the middle budget, with twenty seeds per estimator and exact changed targets (Appendix E). For ACS logistic B8, the mean exact-target shift from B32 is .4977 normalized L1, interval $[.4335, .5641]$, although both estimators have observed success 1 against that new exact target. This is conditional target sensitivity: B32 is not population truth, and one draw per size confounds size with row resampling. Its 144 references and 5,760 calls remain outside the primary repetition estimand.

Actual computation and interrupted elapsed clocks. Main jobs used 7.493 CPU hours: 5.659 for exact references, 1.779 for stochastic calls and .0556 for the remaining process overhead. The pre-run main projection was 7.405 CPU hours; pilots, statistical postprocessing and the separate background arm are outside this measured main total. Peak main process RSS was 831,824 KiB. Across named conditions, mean one-run CPU costs range from .000638 to .339 seconds and mean prediction work from about 911 to 65,569 model-input rows. These are condition means, not service-latency limits or individual maxima. Exact references dominate study cost and are not needed for every production explanation once an assessment has been performed.

Across the disjoint recorded main, preliminary, development and background intervals, the measured subtotal is 7.835 CPU hours. Several setup, failed-control, validation and postprocessing intervals lack CPU timers; their unknown costs are not zero, and this subtotal is not an exact whole-study total (Appendix B).

Equal nominal budgets do not execute equal work: paired integer paths, within-run caching and endpoint/background expansion change prediction rows and CPU costs. The supplied paired nominal-budget contrasts therefore cannot establish equal-cost superiority. Recorded wall times are preserved, including severe interruptions: two German reference calls each exceed 162,000 elapsed seconds while using under 180 CPU seconds. Process CPU and actual rows are the primary efficiency currencies; these interrupted wall clocks describe this execution and cannot serve as a clean deployment-latency benchmark. Overlapping cell elapsed times are not added into a study duration.

7 Discussion

A single stochastic attribution is conditionally adequate for the explicit numerical reading tested here. At the highest budgets it always met the broad magnitude/two-of-three signed-selection criterion in the observed panels; lower budgets, particular fitted predictors and stricter feature-list requirements produced failures. This supports budgeting and checking explanations against the reading they must preserve. It does not support trusting every single run because SHAP satisfies an additive identity, because another run looks similar, or because the panel average is high.

An assessment should specify the probability game and feature units, audit a feasible reference for that game, measure both vector and signed-selection error, and show the distributions across inputs. Work and CPU time make its resource requirement checkable. The changed-background arm further shows why accurate estimation for one chosen empirical game cannot substitute for justifying that game. If the application requires all three listed signed features to be recovered, our broad event is insufficient. If it requires a population or per-prediction reliability certificate, neither the finite observed all-success panels nor thirty seeds supply one. The measured failure cases and inactive low-signal sensitivity also limit any attempt to transfer these numerical tolerances directly to new inputs or tasks.

The contribution is this executed operational assessment and its reproducible evidence, not a new estimator or a universal tolerance. The comparison extends established estimator-error and convergence work to a fully enumerated, fixed-game tabular panel with explicit one-run criteria, matched input uncertainty, separate repeatability, actual computation accounting and retained negative controls. Numerical reliability remains only one requirement for using an attribution as a reason for a consequential prediction.

Broader impacts. Showing when a displayed attribution changes can help users avoid overconfident numerical explanations and make computation requirements transparent. Conversely, an apparently stable feature list can lend unjustified authority to a biased credit or health prediction; even perfect agreement with our empirical game would not make that decision fair or appropriate. We use existing public benchmarks without contacting or recruiting participants, do not redistribute raw person records or release fitted decision services, and retain source-use qualifications and demographic/survey limitations. The study's modest CPU use is reported rather than converted into an unsupported emissions estimate.

8 Scope and limitations of the assessment

The experiment isolates explainer randomness for fixed trained classifiers. A fixed split and one model-training seed do not measure variability from retraining, hyperparameter selection or distribution shift. Three tabular tasks and the selected model families cannot establish a universal ranking of SHAP implementations. In particular, deterministic exact algorithms and newer stochastic estimators have different properties from the two approximations compared here. The logistic classifier is explained on its probability scale, so its linear predictor does not make these probability-game attributions analytically linear.

Exact enumeration removes coalition-sampling error from the selected finite-background reference, subject to floating-point accuracy and validated implementation. It does not make that empirical background representative or resolve the choice between marginal and observational games. Mixing feature blocks can produce unlikely records. A separately changed background defines a different target; its shift must be distinguished from estimation error for either target. The cost of enumeration

grows exponentially with the number of original-feature players. Its feasibility for these modest-dimensional tasks does not supply a general reference strategy for high-dimensional deployment.

The selected magnitude floor and signed top-three criterion encode a numerical reading of an attribution, not a validated human decision rule. Other tolerances can change the adequacy rate, and small reference magnitudes or ties can make a feature list ambiguous. An input-panel mean does not certify each prediction, and thirty seeds cannot rule out rare failures for one input. Pointwise cluster intervals describe sampled-input variability under the stated independence assumption; they are not simultaneous bounds over every comparison. The conservative bounded-mean guard further limits any population reliability certificate, even when no failures are observed.

The data also constrain interpretation. The ACS benchmark is unweighted and samples person rows; original household or survey-design dependence is not modeled by the input bootstrap. The historical Diabetes data concern a narrow demographic sample. No result of this experiment establishes population fairness, causal validity, clinical performance or lending-policy suitability. More reproducible numerical attributions can still give a misleading rationale for a biased or poorly specified model. Reporting the numerical endpoint as a deployment endorsement would exceed the study's evidence.

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A Initial pilot computation accounting

Figure 3 and Table 1 document the initial implementation pilot. They expose the distinction between requested coalition-work settings, actual model-input rows and measured call cost. They are not the main reliability evaluation. The source records, panel/cell mappings and unchanged executable presentation generator accompany the artifacts; these calls must remain separate from the main-study inputs and repetitions.

The pilot's smaller games reached exhaustive Kernel SHAP evaluation at the higher requested setting. Complete coverage of a permutation estimator's coalition cache has a different meaning: random path frequencies can still vary. These observations motivate recording actual work and effective-player exhaustion rather than equating a requested setting with a stochastic sample size or a fixed latency. They do not establish an equal-cost accuracy advantage for either estimator.

These initial timings precede the analysis owner's correction to use float64 prediction inputs consistently across all estimators and references. Subsequent timing and full-study accounting must use that corrected implementation; the raw times here describe the archived pilot only. The initial prototype's reliability metrics are not used to estimate the main operational endpoint.

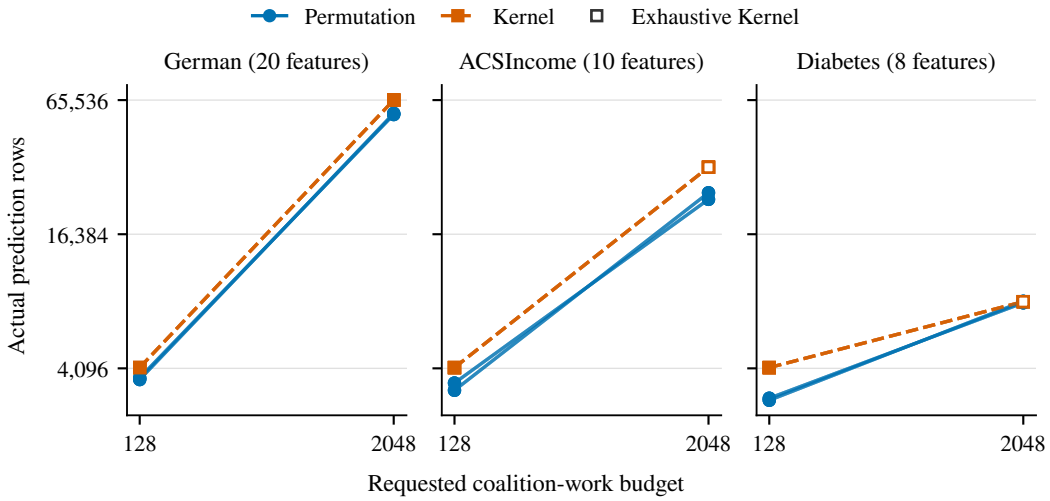


Figure 3: Pilot budget accounting, not main-study reliability evidence. Each line joins two measured calls for one of two pilot inputs per fitted model, with a fixed 32-row empirical background and grouped raw-feature players. The horizontal axis is the requested coalition-work setting; the vertical axis counts actual input rows evaluated by the prediction pipeline. Colors, markers and line styles identify the estimators. Hollow Kernel SHAP markers denote exhaustive evaluation over the effective varying players; complete permutation-cache coverage does not imply an exact path-sampling estimate. Overlapping calls may obscure one another. Traces for the three fitted models coincide in prediction-row count; the accompanying table shows their distinct CPU costs. These pilot calls have no population uncertainty interval. They motivate measuring actual work and using dimension-specific stochastic budgets in the subsequent study.

Table 1: Measured pilot CPU time per explanation call (milliseconds, three significant figures). C1 and C2 are the two pilot inputs per fitted model; each number is one timed call, not an average or confidence interval. Column labels are requested coalition-work settings, not actual prediction-row counts. The background contains 32 rows. Library import was warmed before timing; training and references are excluded. The initial 2048 Kernel setting is exhaustive for ACSIncome and Diabetes and is retained as a labeled control, not stochastic main-study evidence. LR, RF and HGB denote logistic regression, random forest and histogram gradient boosting. This table describes the pilot, not final-study costs.

			Permutation (ms)		Kernel (ms)	
Dataset	Model	Input	128	2048	128	2048
German	LR	C1	2.24	35.0	15.4	163
		C2	1.57	23.8	10.6	132
	RF	C1	24.2	291	47.7	280
		C2	22.9	291	47.1	290
	HGB	C1	19.7	307	32.6	362
		C2	20.9	299	29.1	344
ACSIncome	LR	C1	31.1	223	67.8	516
		C2	29.9	203	73.1	532
	RF	C1	46.4	235	78.5	284
		C2	40.4	253	79.7	295
	HGB	C1	16.4	106	27.8	154
		C2	16.8	129	26.1	137
Diabetes	LR	C1	1.13	4.90	5.50	5.28
		C2	0.930	4.46	5.08	5.28
	RF	C1	20.5	39.3	40.1	51.1
		C2	19.7	41.0	46.6	59.6
	HGB	C1	17.6	51.8	31.7	53.4
		C2	15.2	42.8	29.6	46.9

B Reproducibility and source assets

Delivered implementation. The delivery root contains the report source, built `main.pdf`, analysis package, configurations and presentation scripts. The commands below reproduce the frozen main pipeline; the completed raw checkpoints and numerical summaries are supplied in this delivery. The current evidence status is recorded in `results/main_execution_manifest.json`. That manifest binds the executable configuration and attribution sources to their SHA-256 identity. The complete inventory comprises 540 reference games and 97,200 stochastic calls. Main metadata retain raw-data hashes, source/configuration signatures, seeds, selected input/background IDs, preprocessing groups, library versions and hardware. Checkpoints are reused only when their configuration and source identity match. The independent panel audit checks saved attribution/reference vectors and complete run denominators before aggregation.

Environment and reproduction commands. The analysis uses Python 3.11, with exact top-level packages in `analysis/requirements.txt` and installed versions in `analysis/environment-versions.json`. The analysis package specifies single-threaded BLAS/OpenMP execution and no GPU. From the delivery root, create a Python environment, install the pinned packages, then run:

```
python3 -m venv .venv
. .venv/bin/activate
python3 -m pip install -r analysis/requirements.txt
cd analysis
export OPENBLAS_NUM_THREADS=1
export OMP_NUM_THREADS=1
export MKL_NUM_THREADS=1
python3 scripts/fetch_data.py
PYTHONPATH=src python3 tests/test_estimators.py
PYTHONPATH=src python3 tests/test_metrics.py
python3 scripts/run_queue.py \
  german:logistic german:forest german:histboost \
  acs:logistic acs:forest acs:histboost \
  diabetes:logistic diabetes:forest diabetes:histboost \
  --output ../results/reproduction/main
python3 scripts/audit_panel.py \
  --input ../results/reproduction/main \
  --output ../results/reproduction/panel_audit.json
python3 scripts/aggregate.py \
  --input ../results/reproduction/main \
  --output ../results/reproduction
python3 scripts/evidence_handoff.py \
  --input ../results/reproduction
python3 scripts/audit_timing.py \
  --input ../results/reproduction
```

To regenerate explanations, use a fresh output root: the delivered checkpoints omit fitted joblib bundles. The acquisition script checks the recorded dataset hashes; it does not silently substitute changed data. Separate cell queues can be run in parallel when memory permits. Actual elapsed time depends on that scheduling and on shared-host contention. The accepted pre-execution ledger projected approximately 7.4 CPU hours for the main matrix, or 13.3 CPU hours including planning allowances and reserves, against the objective's approximate 24-CPU-hour planning scale. The measured main total is 7.493 CPU hours on Intel Xeon Gold 6230 CPUs at 2.10 GHz in Linux containers with two-core affinity and one-thread numerical libraries. Reproduction should allow roughly eight CPU hours for the main matrix plus acquisition, preliminary checks, statistical postprocessing and any separately executed sensitivity arm; scheduling determines elapsed duration. The recorded original wall clocks include severe interruptions and are preserved in `results/timing_audit.json`, rather than used as a clean runtime prediction.

To audit and reaggregate the supplied evidence directly, use the same audit/aggregate commands with input `../results/main` and outputs `../results/`, without running acquisition or

model fitting. The fresh-root reproduction preflight independently regenerated one Diabetes logistic main input and all 180 calls with exactly matching saved reference/attribution vectors (`results/reproduction_preflight.json`); it is not an additional statistical replication. For the separate background arm, after fresh main reproduction run:

```
python3 scripts/run_background_queue.py \
  --main-results ../results/reproduction \
  --output ../results/reproduction/background
python3 scripts/aggregate_background.py \
  --main-results ../results/reproduction \
  --input ../results/reproduction/background \
  --output ../results/reproduction/background_sensitivity
```

Audit the delivered background checkpoints instead with main-results `../results`, input `../results/background` and output `../results/background_sensitivity`. The shipped main and background raw vectors are already present; reaggregation requires no new explanations.

Whole-study computation accounting. `results/computation_ledger.json` and its CSV record the known and uninstrumented executed stages, source hashes and deduplication rules. Disjoint process-clock intervals sum to 28,206.959 seconds (7.835266 CPU hours), including the 7.49345-hour main jobs, four distinct preliminary timing stages, ACS preflight, three development smokes, a fresh reproduction preflight and the background pilot/full phase timers. This is an incomplete recorded subtotal. The four preliminary stages together have 708.346 recorded CPU seconds; their startup/loading/writing outside those clocks is unknown. The unsuccessful TreeExplainer strict abort/diagnostics, early precision/parser corrections, final common-game validation and repeated local tests/audits/postprocessing retain explicit untimed entries. None is assigned zero or estimated from wall time.

Managed resource observations are separate, sampled and nonadditive: the archived cumulative counter reports 7.855328 CPU hours, and managed main/background postprocessing observations are 46.904/2.965 seconds. These overlap process clocks, omit sandbox-only work and can undershoot a phase clock locally. Neither adding the counter to the subtotal nor calling either an exact all-inclusive cost is justified. The supplied approximate 24-CPU-hour scale was a planning estimate; absent instrumentation limits the achieved total we can establish. `python3 scripts/computation_ledger.py -input ../results`, run from `analysis/`, regenerates the ledger from preserved source records without rerunning experiments.

Predictor details. The fixed model configuration is logistic regression with $C = 1$ and at most 2,000 iterations; a random forest with 100 trees, minimum leaf size 5 and a feature fraction of .8; and histogram gradient boosting with 100 iterations, at most 15 leaves, L2 regularization 1 and early stopping disabled. Other constructor defaults are fixed by scikit-learn 1.9.1. These configurations were selected for contrasting CPU-feasible predictors rather than tuned to explanation quality. Numeric preprocessing is fitted per raw feature using training-only median imputation and standardization. Categorical preprocessing uses training-only most-frequent imputation and either one-hot encoding for logistic regression or ordinal codes for trees; unknown categories are ignored by the former and mapped to -1 by the latter. The original-feature group map is retained through the transformation. Train/test splitting and model training use seed 20261103, capped training subsampling uses seed 20261104, and background/panel selection uses seed 20261105. Explainer seeds and bootstrap seeds are stated in Section 5 and retained in run metadata.

German Credit. We use Hofmann's Statlog German Credit data from UCI, DOI 10.24432/C5NC77, distributed under CC BY 4.0 [Hofmann, 1994]. The original symbolic `german.data` supplies the 20 feature players, rather than the separately recoded numeric representation. The positive class is recoded from original label 2; splitting and train-only preprocessing are study transformations. Attribution to the source does not imply its endorsement of these analyses.

ACSIncome. The source is the U.S. Census Bureau's ACS 2018 1-Year Public Use Microdata Sample, California person records, `csv_pca.zip`, retrieved on 3 November 2026 [U.S. Census Bureau, 2018]. ACSIncome is derived with folktables 0.0.12 following Ding et al. [2021]; the filter

and weighting policy are stated in Section 5. The archive is obtained from the documented file route over HTTPS, not through the Census API. The checked Census guidance provides public-use and citation statements; no separate named license for this archive was located on the checked official pages. The MIT license for folktables software does not license the Census records. Census is the source of the original data only; these analyses and conclusions are the authors' responsibility and imply no Census endorsement.

Diabetes. We use OpenML data ID 37, version 1 [OpenML, 2014]. Its metadata attribute the source to the National Institute of Diabetes and Digestive and Kidney Diseases and donor Vincent Sigillito, and label the license field *Public*; separate upstream license wording was not verified. This historical sample concerns women of Pima Indian heritage aged at least 21 near Phoenix. The target coding and declared zero-to-missing transformations are retained in the acquisition/loader documentation. This benchmark is not a clinical-validation cohort.

Source and display provenance. `analysis/data/manifest.json` gives retrieval URLs, dates, byte counts and SHA-256 hashes; `analysis/notes/data-attribution.md` and its source snapshots preserve the bounded source-use statements and reading provenance. Raw archives are retrieved upstream rather than redistributed. The initial pilot plot and timing table in Appendix A are generated together by `figures/generate_pilot_accounting.py`; their manifests map every panel and table entry to the unchanged pilot records. They can be regenerated from the delivery root with `python3 figures/generate_pilot_accounting.py -preview`. All main and background figure/table assets regenerate with `bash figures/reproduce_main.sh` from the delivery root; per-artifact generator/input mappings are in their manifests and Appendix F. Main numerical inputs are the complete `results/` panel and aggregates; final figure/table commands and per-cell provenance are documented under `figures/` and `tables/`. Build the report with `bash scripts/build_paper.sh`; this requires TeX Live with pdfLaTeX, BibTeX and latexmk, plus Poppler and ripgrep. No source in this delivery was obtained by seeking the deliberately withheld prior study; consulted materials and reading extent are recorded in `paper/source-log.md`.

Research-process disclosure. Language-model agents assisted research design, implementation, interpretation, manuscript drafting and review within separate research roles. They are part of the research process, not an estimator, reference oracle, data source or simulated participant in the empirical study. Every empirical number comes from executed numerical code and retained outputs; source/configuration fingerprints, analytic tests, saved-vector audits and independent revision-specific review constrain claims. No claim of human-subject review or clinical validation follows from this workflow.

C Expanded target and assessment definitions

C.1 Attribution target and estimator diagnostics

A fixed explanation problem. Let f be a trained classifier's probability for a specified class, x a held-out input, and $N = \{1, \dots, d\}$ the set of original input features. A categorical feature remains one player even when its preprocessing expands it into several columns. Let $B = (z^{(1)}, \dots, z^{(m)})$ be a fixed background drawn from the training data. We consider the empirical marginal game

$$v_{x,B}(S) = \frac{1}{m} \sum_{b=1}^{m} f\left(x_S, z^{(b)}_{N \setminus S}\right), \qquad S \subseteq N, \tag{3}$$

where the argument is transformed by the fitted preprocessing pipeline. Replacing an entire original feature preserves its encoding as a group. The Shapley attribution for this game is

$$\phi_j(x; f, B) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(d - |S| - 1)!}{d!} \left[v_{x,B}(S \cup \{j\}) - v_{x,B}(S)\right]. \tag{4}$$

These values allocate $f(x) - v_{x,B}(\emptyset)$ among the features in probability units. The additive framework and sampling approximations are established by Lundberg and Lee [2017]. Exactness below always refers to the selected finite-background game, up to numerical precision; it does not mean that the background equals the population distribution.

Equation (3) averages over background rows without conditioning their missing coordinates on x_S. Observational and interventional value functions answer different questions, and their suitability depends on the application [Chen et al., 2020]. Background mixing can create combinations unlikely under the observed joint distribution. Consequently, numerical accuracy for this game does not establish a causal account, fairness, clinical validity, or the appropriateness of a prediction for deployment.

The random object is the estimator. Write $\widehat{\phi}_{q,r}$ for one approximation using computation setting q and random seed r. Holding f, x, B, the preprocessing, output class and feature players fixed isolates explainer randomness. Changing the fitted model or background changes the conditioning of the question and must be analyzed separately. The cost setting is described by both the requested work and the work actually executed, including model-input rows expanded over the background, together with observed CPU and elapsed time.

Agreement, accuracy, and use. Repeatability compares $\widehat{\phi}_{q,r}$ with an independent run $\widehat{\phi}_{q,r'}$. Estimation accuracy instead compares a run with ϕ or a separately validated reference for the same game. These are distinct assessments: two estimates can agree while sharing error. Likewise, the efficiency residual $\sum_j \widehat{\phi}_j - [f(x) - v_{x,B}(\emptyset)]$ checks the total allocation, not its division among features. A zero residual cannot certify correct coordinates.

Vector precision, top-feature identity, salient-feature membership and contribution signs require different numerical tolerances, especially near zero and near ties. An operational criterion names that reading and resolution; panel averages do not certify each individual prediction.

C.2 Assessment criteria and inference

Magnitude and salient features. For a reference vector $\phi^\star$ for the same game, define absolute and normalized errors as

$$E_1(\widehat{\phi}, \phi^\star) = \|\widehat{\phi} - \phi^\star\|_1, \qquad E_{\mathrm{rel}}(\widehat{\phi}, \phi^\star) = \frac{E_1(\widehat{\phi}, \phi^\star)}{\max(\|\phi^\star\|_1, \delta)}, \tag{5}$$

where $\delta > 0$ is an explicitly declared resolution floor in probability units. Absolute error retains its meaning when the reference has little attribution mass; normalized error expresses the error relative to its scale, subject to that floor. Reporting the reference norm, weak-signal cases, and floor sensitivity prevents a single normalized summary from concealing either small effects or a permissive denominator.

Let $T_k(a)$ select the k largest coordinates of a in absolute magnitude, using a declared deterministic rule for ties. Unsigned and signed overlaps are respectively

$$O_k(\widehat{\phi}, \phi^\star) = \frac{|T_k(\widehat{\phi}) \cap T_k(\phi^\star)|}{k}, \qquad O_k^{\pm}(\widehat{\phi}, \phi^\star) = \frac{1}{k} \sum_{j \in T_k(\widehat{\phi}) \cap T_k(\phi^\star)} \mathbf{1}\{\mathrm{sgn}(\widehat{\phi}_j) = \mathrm{sgn}(\phi_j^\star)\}. \quad (6)$$

The sign describes the direction of contribution to the selected class relative to the game's baseline; it is not the direction of a causal effect. Unsigned overlap separates membership changes from direction changes. Top-1 identity and magnitude-rank agreement provide complementary diagnostics. Tied boundaries and near-zero signs need an ambiguity analysis alongside deterministic point scores.

An explicit one-run tolerance scenario can combine magnitude and signed membership:

$$A_{\tau,\eta}(\widehat{\phi}, \phi^\star) = \mathbf{1}\{E_{\mathrm{rel}} \le \tau \text{ and } O_k^{\pm} \ge \eta\}. \quad (7)$$

The tolerance τ, overlap requirement η, set size k, resolution floor δ, and their sensitivity grids are design choices declared before the affected main runs. Components must be reported separately: the same joint failure can arise from inaccurate magnitudes, unstable membership, or disagreeing signs. Crossing an operational threshold establishes only that particular numerical criterion.

Repeatability and the unit of uncertainty. For each fixed input, independent seeds produce R approximation vectors. Pairwise agreement can use all distinct run pairs, but pairs sharing a run are dependent. Their within-input average is one summary for that input; it does not create a larger independent input sample. A direct magnitude comparison uses the same reference-norm floor for pairwise mean L1 disagreement and run–reference error. A reference-free normalization by the runs' own norms is a separately named diagnostic. Even with a common scale, two-run disagreement is not the error of one run. Likewise, a panel-average one-run adequacy estimate is

$$\widehat{p}_{\tau,\eta} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{R} \sum_{r=1}^{R} A_{\tau,\eta}(\widehat{\phi}_{i,r}, \phi_i^\star). \quad (8)$$

This estimates performance across the stated input panel and seed distribution, conditional on the fitted model, split, empirical background and implementation. It does not average over alternative training runs or backgrounds. Instance-level distributions must accompany the panel mean. Inference over sampled inputs keeps all repetitions of an input together; paired comparisons across estimators or budgets preserve the matched inputs. A finite set of runs with no observed failures does not prove that an input is always explained adequately.

References and ambiguity. An enumerated reference can be exact for the selected empirical game. A high-budget stochastic reference instead requires independent draws and its own numerical assessment. The mean of the low-budget runs being judged is not an independent truth target. If remaining reference uncertainty can change a score, top-feature identity, or tolerance event, that ambiguity must be retained in denominators and reported rather than removed silently. Repeatability remains measurable when target accuracy is unresolved, but it cannot replace that accuracy assessment. Efficiency, reference convergence, actual computation and failed-run status therefore accompany the reliability criteria.

Boundary observations do not certify reliability. For bounded rates and overlaps, a panel whose input-level means are all zero or all one gives a degenerate percentile-bootstrap interval. Such a panel retains its raw bootstrap endpoints, but the displayed interval uses a two-sided Hoeffding bound: $[\widehat{p} - h, \widehat{p} + h] \cap [0, 1]$, where $h = \sqrt{\log(2/.05)/(2n)}$. A separate conservative one-sided lower bound is $\max(0, \widehat{p} - \sqrt{\log(1/.05)/(2n)})$. These bounds assume independent bounded input-level observations; repeated seeds or shared run pairs do not increase n. They do not remedy dependence among original records or establish survey-population coverage. Interior percentile-bootstrap intervals remain descriptive approximations. With 60 inputs, even an observed all-success panel cannot clear a population success threshold of .95 using this conservative guard. This limits a certification claim; it is neither an observed failure of the explainer nor a reason to change the adequacy threshold. Per-input intervals over the 30 seeds describe that input separately and do not certify every input's performance.

D Implementation checks and an unsuccessful control

These are completed development checks, separate from the main held-out reliability panel. Their raw records and executable checks are delivered with the analysis. They support the choice and numerical implementation of the reference; they do not establish main-panel adequacy, reference completeness or achieved runtime.

Common-game numerical check. The original prediction adapter could change low-order probability digits with batch shape when its inputs remained float32. The corrected adapter casts prediction inputs and probability accumulation to float64 for both approximations and direct enumeration. In `results/common_game_validation.json`, exhaustive Kernel SHAP and direct enumeration are compared on two pilot inputs for each of three model types on ACSIncome and Diabetes. All twelve comparisons pass the declared 10^{-12} maximum-coordinate-error check; the largest observed discrepancy is 4.44×10^{-16} probability units. The generating script is `analysis/scripts/validate_common_game.py`. This checks two computational routes for the selected pilot games, alongside the analytic estimator tests; the completed main reference inventory separately passes its saved-vector audit (Section 6).

Exploratory TreeExplainer control. An attempted additional random-forest check used TreeExplainer with interventional probability output against archived direct-enumeration pilot references. All five checked inputs failed the declared 10^{-6} coordinate-agreement criterion. The observed maximum-coordinate discrepancies range from 1.06×10^{-4} to 3.73×10^{-4} probability units, and the TreeExplainer allocations also have nonzero efficiency residuals. Full vectors, baselines, residuals and failure flags are retained in `results/tree_control_validation.json`; `analysis/scripts/validate_tree_control.py` produces the diagnostic record. The initial strict check stopped on failure; the diagnostic version retains the unsuccessful comparisons. This control was not adopted as a reference. Its numerical or version compatibility was not resolved, so these observations support neither an independent validation by TreeExplainer nor a general claim about Tree SHAP accuracy. Direct enumeration remains the reference route for every main game.

E Separate changed-background sensitivity

Design and target. After the primary fixed-background panel, a bounded supplementary arm used ACSIncome and Diabetes with all three previously fitted models. It retained the first twelve main inputs in their original random order and did not refit models or redraw explained inputs. For each fitted cell, one background of 8 and one of 128 training rows were drawn without replacement, with recorded SeedSequence inputs beginning with 60793. Each game was exactly enumerated, and each estimator ran twenty independent seeds at the dataset's middle nominal budget (ACS 256; Diabetes 64). Seeds, training/background/source-input IDs, signatures, vectors and costs are retained in `results/background/`. The arm contains 144 changed-game exact references and 5,760 stochastic calls across six fitted cells, separately audited and aggregated under `results/background_sensitivity/`.

Two comparisons. Estimation accuracy compares a stochastic vector with the exact reference for its own changed background, using the same primary criterion and its new reference norm. Target sensitivity instead compares that changed exact vector with the original B32 exact vector on the same input. For the latter comparison, normalized L1 uses the B32 norm and floor; B32 is a comparison target, not population truth. The stored reference-shift columns named error or match describe differences between two exact games. These are not causal errors or extra primary estimator replicates. Instance-cluster uncertainty uses twelve inputs, preserving repeats for the stochastic comparison; all-success displayed intervals are $[.6080, 1]$, rather than zero-width bootstrap endpoints.

Measured target shifts and stochastic accuracy. For B8, mean normalized exact-target shifts from B32 range from $.2914$ for ACS forest to $.5364$ for Diabetes histogram boosting; for B128 they range from $.1247$ to $.2606$ on those same fitted cells. ACS logistic B8 has mean shift $.4977$, interval $[.4335, .5641]$, while both stochastic estimators have observed success 1 against that B8 target. The most salient exact feature changes on 4/12 of those inputs: top-one agreement $.6667$, interval $[.4167, .9167]$. Thus high estimation accuracy for one game does not establish invariance to a different background choice.

Across changed-game conditions, observed primary success ranges from $.9083$ to 1. The lowest condition is Diabetes histogram boosting with permutation at B8: $.9083$ $[.8125, .9792]$. B128 condition means are $.9958$ or 1. Full signed-selection, absolute-shift and top-one outcomes are supplied, with all twelve cases retained. Only one row sample per size is used, so size and resampling effects are confounded. This arm supports conditional target sensitivity for the tested draws, not a pure-size effect, population background variance, German Credit generalization or model-retraining robustness.

Cost and software notices. The timed pilot preceded expansion, with 96 stochastic calls and 24 exact targets, and projected 17.44 CPU minutes (26.16 with its safety allowance). Full supplementary cell-phase timers record 466.053 CPU seconds, excluding initial imports and bundle/data loading. Sampled resource-reported job costs are retained separately and can undershoot these phase timers; neither scope is an exact all-inclusive CPU total. All 18 captured first-import Matplotlib colormap-deprecation notices are retained with their disposition; no numerical warning or failed attempt was recorded. The background arm does not inherit the primary panel's zero-warning count.

Audit and reproduction. The supplied audit verifies the full source/grid/seeds, training-only background draws, original input identity, main comparison reference, exact endpoints/efficiency and Kernel design rank. Its seeds are unique within the supplementary arm and disjoint from primary seeds. Audit/reaggregation uses the shipped checkpoints without a fitted bundle. To rerun explanations, first regenerate main models into a fresh output root, then run `analysis/scripts/run_background_queue.py` against that root and `analysis/scripts/aggregate_background.py`; exact integrated commands are in `analysis/README.md`. The raw main checkpoints intentionally omit fitted joblib bundles, so requesting resume against them is not the regeneration route.

Table 2: Completed changed-background phase costs: each row has 24 exact targets and 960 stochastic calls (12 inputs, two background draws, two estimators and 20 repeats). Phase CPU/wall starts after imports, bundle and data loading, so it is not comparable to the full-process main total without the scope difference. Peak RSS is whole process. Reference and call timing scopes overlap these phase totals and must not be added. The complete arm has 144 references, 5760 calls and no failed attempts. Its 18 captured first-import colormap-deprecation messages are retained in warning_disposition.json; no numerical warning was recorded. One background draw per size does not estimate a pure-size effect.

Dataset	Model	Phase CPU s	Phase wall s	Peak MiB
ACSIncome	LR	165.06	165.12	710
ACSIncome	RF	155.82	155.86	391
ACSIncome	HGB	77.60	77.61	367
Diabetes	LR	4.80	4.80	310
Diabetes	RF	34.69	34.71	311
Diabetes	HGB	28.09	28.10	310

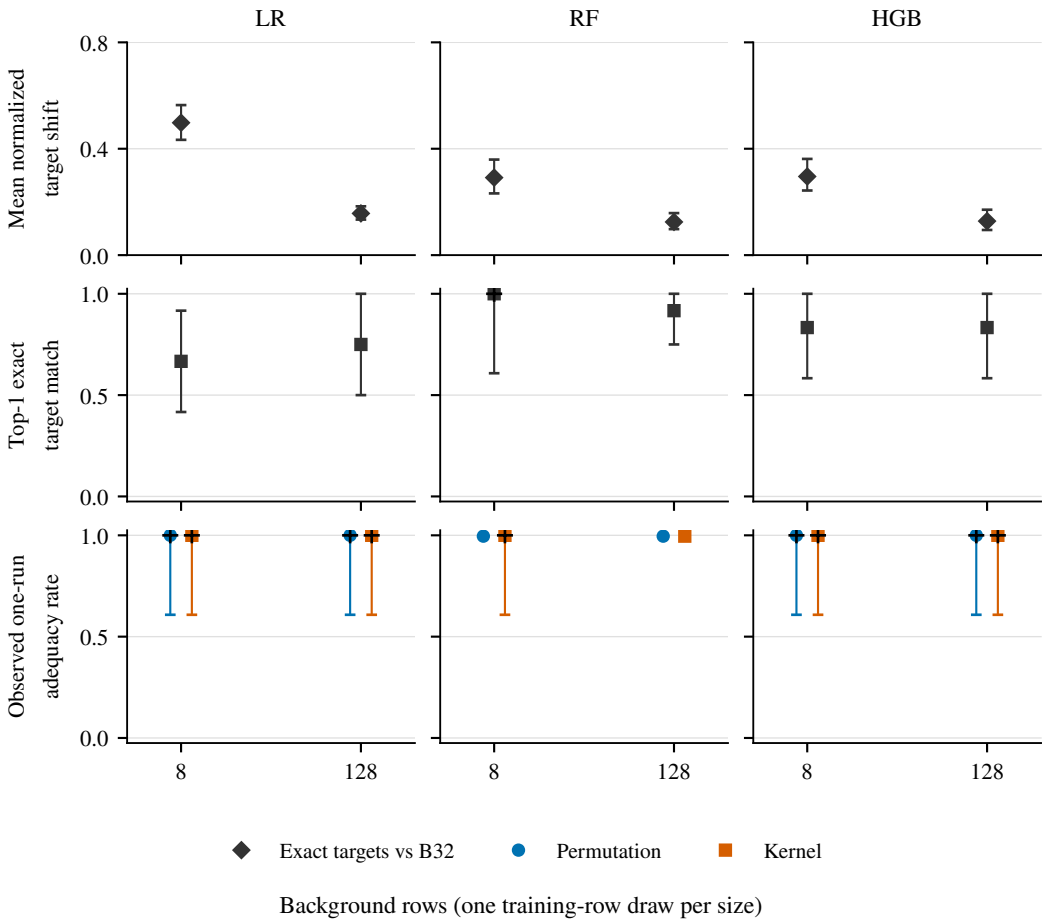


Figure 4: Changed-background sensitivity for ACSIncome. Top: exact changed-game attribution versus the main B32 exact attribution, using L1 difference/max(B32 attribution L1,0.01). Middle: fraction of inputs with the same most salient absolute-attribution feature under those two exact targets (stable raw-index tie order). These measure a target shift, not estimator error; B32 is a comparison target, not population truth. Bottom: stochastic accuracy against each changed game's own exact target, requiring normalized L1 <=0.20 and signed top-3 overlap >=2/3 with floor 0.01. Both estimators use the middle requested budget (256); actual work is not equated. The same 12 main inputs and fitted models are retained, with 20 independent runs per input/estimator/background; exact target comparisons use one value per input. Bars are supplied pointwise 95% instance-cluster percentile intervals, conditional on each fitted model and each single background draw; black plus marks identify supplied two-sided Hoeffding boundary intervals for all-success panels. No >=0.95 population or individual certificate follows. Whole-row empirical interventional integration uses positive-class probability and grouped raw features. Size and selected training rows change together; one draw per size cannot identify a pure-size effect or background-population variance. These calls are separate from fixed-B32 main repetitions.

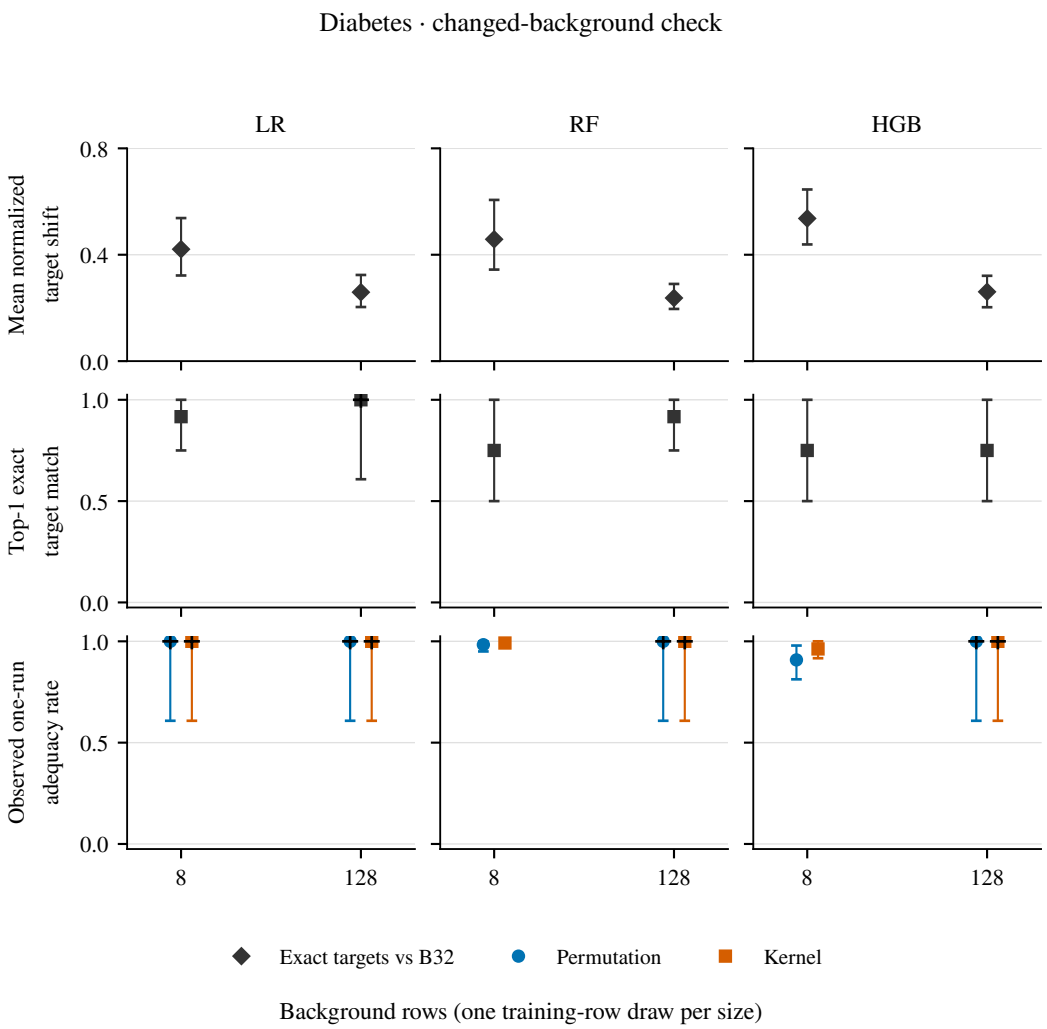


Figure 5: Changed-background sensitivity for Diabetes. Top: exact changed-game attribution versus the main B32 exact attribution, using L1 difference/max(B32 attribution L1,0.01). Middle: fraction of inputs with the same most salient absolute-attribution feature under those two exact targets (stable raw-index tie order). These measure a target shift, not estimator error; B32 is a comparison target, not population truth. Bottom: stochastic accuracy against each changed game's own exact target, requiring normalized L1 <=0.20 and signed top-3 overlap >=2/3 with floor 0.01. Both estimators use the middle requested budget (64); actual work is not equated. The same 12 main inputs and fitted models are retained, with 20 independent runs per input/estimator/background; exact target comparisons use one value per input. Bars are supplied pointwise 95% instance-cluster percentile intervals, conditional on each fitted model and each single background draw; black plus marks identify supplied two-sided Hoeffding boundary intervals for all-success panels. No >=0.95 population or individual certificate follows. Whole-row empirical interventional integration uses positive-class probability and grouped raw features. Size and selected training rows change together; one draw per size cannot identify a pure-size effect or background-population variance. These calls are separate from fixed-B32 main repetitions.

F Complete main numerical displays

These displays retain the complete fixed-background condition grid, its actual work/CPU, predictive coverage, ambiguity counts, component/tolerance sensitivities and input distributions. All plotted intervals are supplied by the analysis; no new inference is computed by presentation. Their manifests map panels and table cells to the analysis records. Reproduce every main numerical asset from the delivery root with `bash figures/reproduce_main.sh`; `figures/MAIN_HANDOFF.md` and `tables/README.md` document each generator, input and artifact. The main text carries the adequacy/work and compact criterion/input displays. The supplementary views below support checking all reported conditions and limitations.

Table 3: The complete handoff contains 9 fitted-model cells, 60 non-pilot explained inputs per cell, 30 independent runs per estimator/budget/input and a fixed 32-row training background. Predictive quality is measured on the held-out split; features are grouped raw players. LR, RF and HGB denote logistic regression, random forest and histogram gradient boosting. These fitted-model diagnostics do not estimate attribution reliability.

Dataset	Model	Train	Test	Features	AUROC	Accuracy
German	LR	800	200	20	0.852	0.805
German	RF	800	200	20	0.825	0.800
German	HGB	800	200	20	0.823	0.810
ACSIncome	LR	20000	39133	10	0.893	0.816
ACSIncome	RF	20000	39133	10	0.885	0.808
ACSIncome	HGB	20000	39133	10	0.896	0.815
Diabetes	LR	614	154	8	0.894	0.792
Diabetes	RF	614	154	8	0.893	0.805
Diabetes	HGB	614	154	8	0.879	0.812

Table 4: All lowest-requested-budget conditions. Adequacy means L1/max(exact-reference L1,0.01) ≤ 0.20 and signed top-3 overlap $\geq 2/3$ for the fixed B32 empirical interventional probability game over raw-feature groups. Rates average 60 input clusters with 30 independent runs each; brackets are supplied pointwise 95% instance-cluster percentile intervals. † marks an observed all-success panel with the supplied two-sided Hoeffding boundary interval instead. Actual rows and process CPU milliseconds are means per call; requested budget labels do not equate realized work. At every highest budget (German 2048, ACS 512, Diabetes 128), observed primary rates are 1 with displayed interval [0.825,1]; none certifies population adequacy ≥ 0.95. Intervals condition on fixed models/split/background.

Dataset	Model	Method	Budget	Rows	CPU ms	Rate [95% interval]
German	LR	Permutation	128	3,668	1.79	$1.000\ [0.825, 1.000]$†
German	LR	Kernel	128	4,129	12.39	$1.000\ [0.825, 1.000]$†
German	RF	Permutation	128	3,669	23.65	$0.997\ [0.992, 1.000]$
German	RF	Kernel	128	4,129	46.52	$0.998\ [0.995, 1.000]$
German	HGB	Permutation	128	3,670	19.92	$0.979\ [0.963, 0.992]$
German	HGB	Kernel	128	4,129	30.09	$0.949\ [0.913, 0.980]$
ACSIncome	LR	Permutation	64	1,694	11.24	$0.999\ [0.997, 1.000]$
ACSIncome	LR	Kernel	64	2,081	36.27	$0.959\ [0.920, 0.988]$
ACSIncome	RF	Permutation	64	1,690	28.73	$0.944\ [0.902, 0.977]$
ACSIncome	RF	Kernel	64	2,081	55.48	$0.854\ [0.777, 0.923]$
ACSIncome	HGB	Permutation	64	1,694	10.01	$1.000\ [0.825, 1.000]$†
ACSIncome	HGB	Kernel	64	2,081	17.76	$0.902\ [0.843, 0.953]$
Diabetes	LR	Permutation	32	915	0.64	$1.000\ [0.825, 1.000]$†
Diabetes	LR	Kernel	32	1,057	3.77	$0.999\ [0.998, 1.000]$
Diabetes	RF	Permutation	32	911	13.61	$0.995\ [0.989, 0.999]$
Diabetes	RF	Kernel	32	1,057	32.93	$0.901\ [0.863, 0.934]$
Diabetes	HGB	Permutation	32	912	6.97	$0.926\ [0.895, 0.953]$
Diabetes	HGB	Kernel	32	1,057	12.98	$0.779\ [0.722, 0.834]$

Table 5: The complete handoff contains 9 fitted-model cells, 60 non-pilot explained inputs per cell, 30 independent runs per estimator/budget/input and a fixed 32-row training background. Reference-defined caution counts; counts are per input and are identical across the method/budget cells, not summed across them. All exact-reference inputs remain in the primary event denominator, including ties and weak signals. Numerical top-1 and top-3-boundary ties use the supplied twice-coordinate tolerance; selected near-zero signs use the coordinate tolerance. Weak signal means reference L1 below 0.01 probability units; active-sign counts identify inputs with at least one active sign. The declared coordinate tolerance is 1e-06 probability units. No unresolved reference or failed final call is present in this complete handoff.

Dataset	Model	Exact	Top-1 ties	Top-3 ties	Near zero	Weak	Active
German	LR	60	0	0	0	0	60
German	RF	60	0	0	0	0	60
German	HGB	60	0	0	0	0	60
ACSIncome	LR	60	0	0	0	0	60
ACSIncome	RF	60	0	0	0	0	60
ACSIncome	HGB	60	0	0	0	0	60
Diabetes	LR	60	0	0	0	0	60
Diabetes	RF	60	0	0	0	0	60
Diabetes	HGB	60	0	0	0	0	60

Table 6: Measured completed-job costs, separate from the initial pilot and resource projections. CPU and wall columns give the successful completion process time; reference CPU sums all saved exact-reference call timings. These scopes overlap and must not be added. A resumed process can exclude earlier work, which belongs in the full research compute ledger alongside pilots and failed attempts. Peak memory is the completion-process peak RSS. The independent saved-panel audit records 0 failed attempts, 0 warning messages and 0 exhaustive calls among the nominally stochastic runs; those counts must accompany any final efficiency or stochastic-method claim. Severe execution interruptions contaminate wall times; they are preserved execution records, not deployment-latency benchmarks. Use CPU and actual rows for efficiency comparisons. Runtime excludes untracked organizational work and is not a per-call latency quantile.

Dataset	Model	CPU min	Wall min	Ref. CPU min	Peak MiB
German	LR	24.63	29.32	17.589	499
German	RF	175.80	2905.87	151.285	521
German	HGB	195.78	2926.88	169.846	486
ACSIncome	LR	19.04	19.07	0.299	812
ACSIncome	RF	16.83	21.55	0.285	436
ACSIncome	HGB	8.06	8.07	0.158	436
Diabetes	LR	0.78	0.79	0.001	314
Diabetes	RF	5.35	5.36	0.035	314
Diabetes	HGB	3.33	3.35	0.045	314

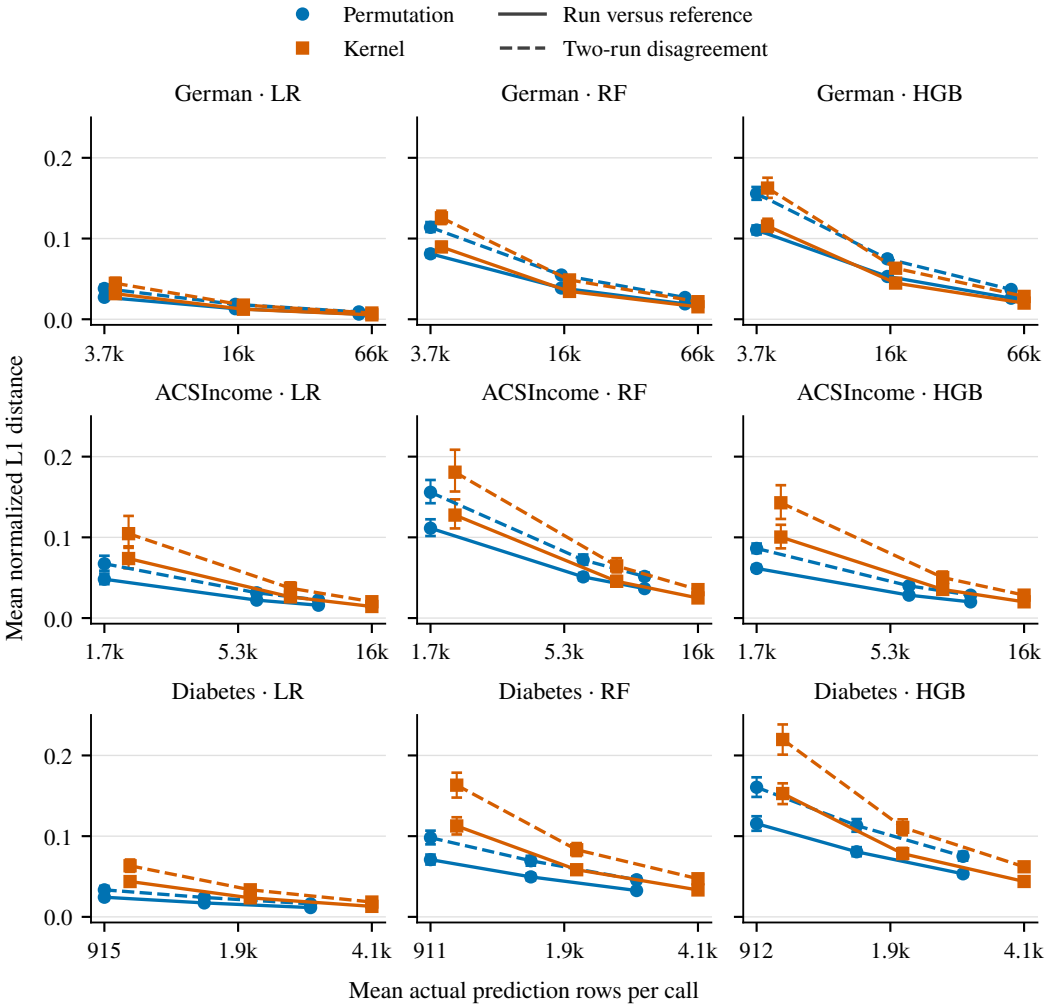


Figure 6: Run-reference error and two-run disagreement use the same denominator, max(exact-reference L1,0.01 probability units). Run-reference distances average 30 runs per input; two-run distances average all distinct run pairs once per input, then equally over 60 independent inputs. Supplied pointwise 95% instance-cluster percentile intervals are conditional on the fitted model, split and background. Two-run disagreement is not single-run accuracy; equality of the curves does not validate an explanation. Horizontal positions are mean measured prediction rows per single call. Both methods target the same empirical interventional probability game over grouped raw features with B=32.

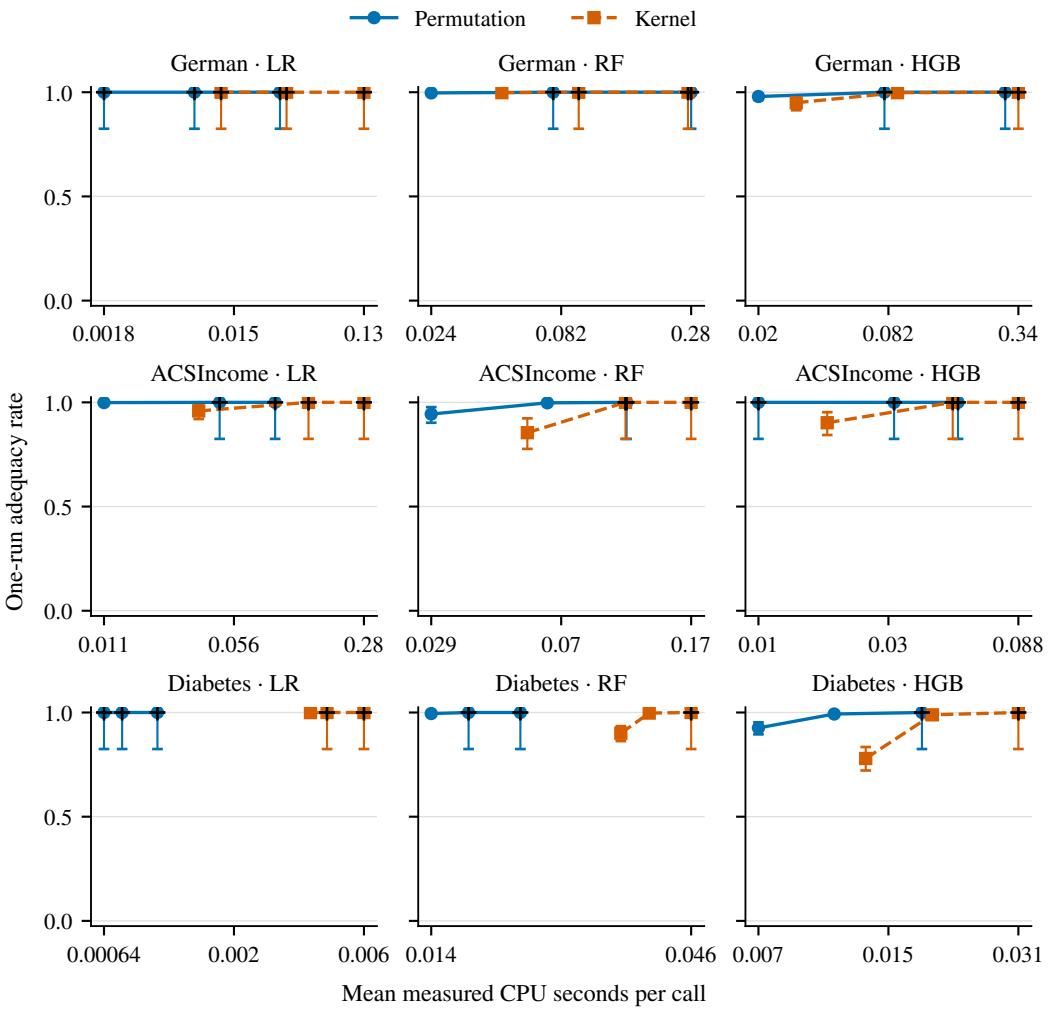


Figure 7: One-run adequacy: normalized L1 error $\leq$0.20 and signed top-3 overlap $\geq$2/3 against the exact finite-game reference. L1 normalization is max(reference L1,0.01 probability units). For each fixed fitted model, 60 held-out inputs and 30 independent runs per input, estimator and requested budget; background B=32. The target is positive-class probability for the fixed empirical interventional game over grouped raw features. Estimates average equally over inputs and runs. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers identify observed all-zero/all-one panels whose displayed bars instead use the supplied two-sided 95% bounded-instance Hoeffding interval. These are observed rates and descriptive intervals, not population 95% adequacy certification. Horizontal positions are mean measured per-call process CPU time, excluding training and reference computation. CPU is the main timing comparison because recorded wall time contains severe execution interruptions.

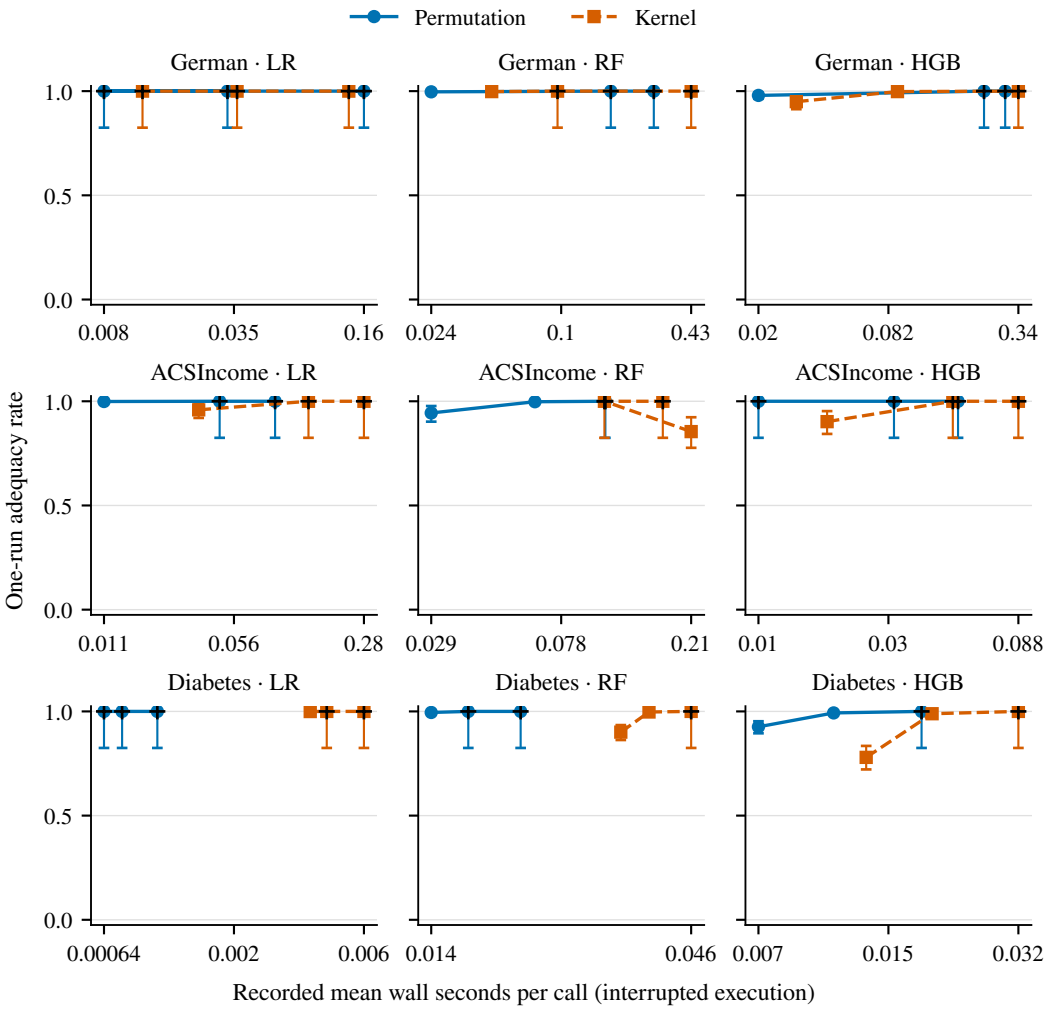


Figure 8: One-run adequacy: normalized L1 error $\leq$0.20 and signed top-3 overlap $\geq$2/3 against the exact finite-game reference. L1 normalization is max(reference L1,0.01 probability units). For each fixed fitted model, 60 held-out inputs and 30 independent runs per input, estimator and requested budget; background B=32. The target is positive-class probability for the fixed empirical interventional game over grouped raw features. Estimates average equally over inputs and runs. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers identify observed all-zero/all-one panels whose displayed bars instead use the supplied two-sided 95% bounded-instance Hoeffding interval. These are observed rates and descriptive intervals, not population 95% adequacy certification. Horizontal positions are mean measured per-call wall time, excluding training and reference computation. Severe execution interruptions and shared-host scheduling are retained. These are execution records, not deployment-latency benchmarks or latency quantiles; use CPU/actual-row curves for efficiency comparison.

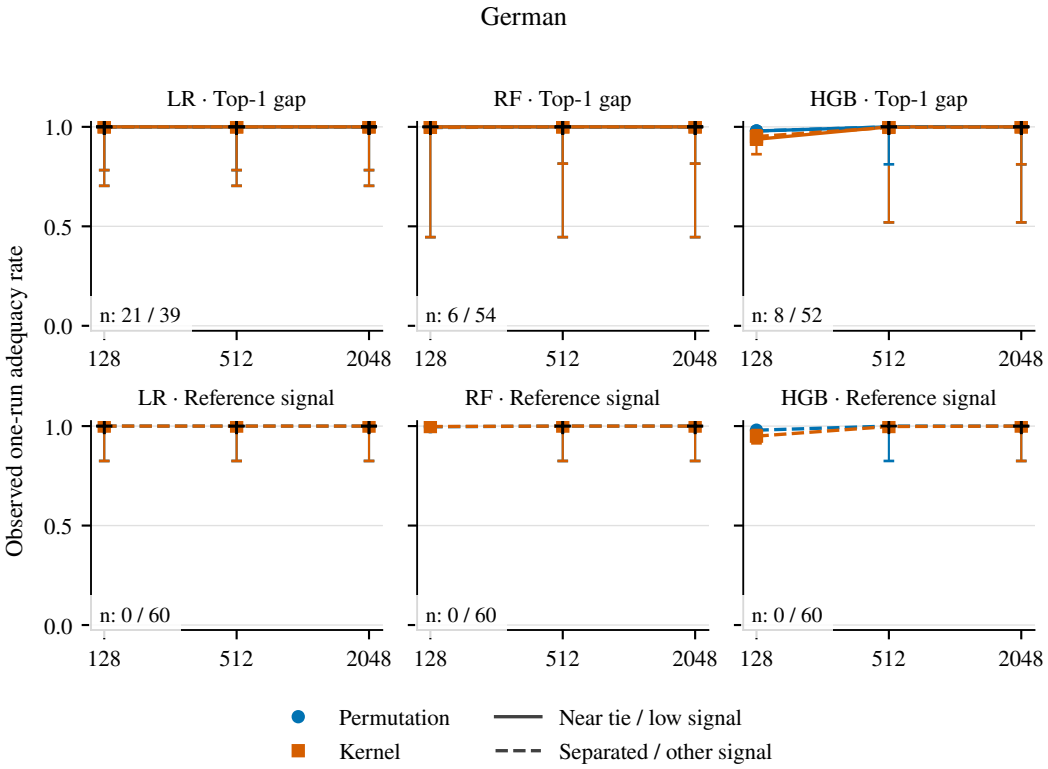


Figure 9: Reference-defined conditions for German. Upper row: near-top-1 tie versus separated, using (largest minus second-largest absolute reference attribution)/max(reference L1,0.01) $\leq$0.02. Lower row: low signal (reference L1 <0.01 probability units) versus other signal. Each panel labels independent-input counts in solid/dashed order; counts do not multiply across runs or budgets. Empty strata have no estimate; singleton strata show the observed estimate without an interval. Adequacy requires normalized L1 $\leq$0.20 and signed top-3 overlap $\geq$2/3. These descriptive strata are fixed by the exact reference, not selected using explanation performance. No main input has low signal, so that empty stratum is untested rather than a zero-success finding. Requested work labels do not compare equal actual work. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

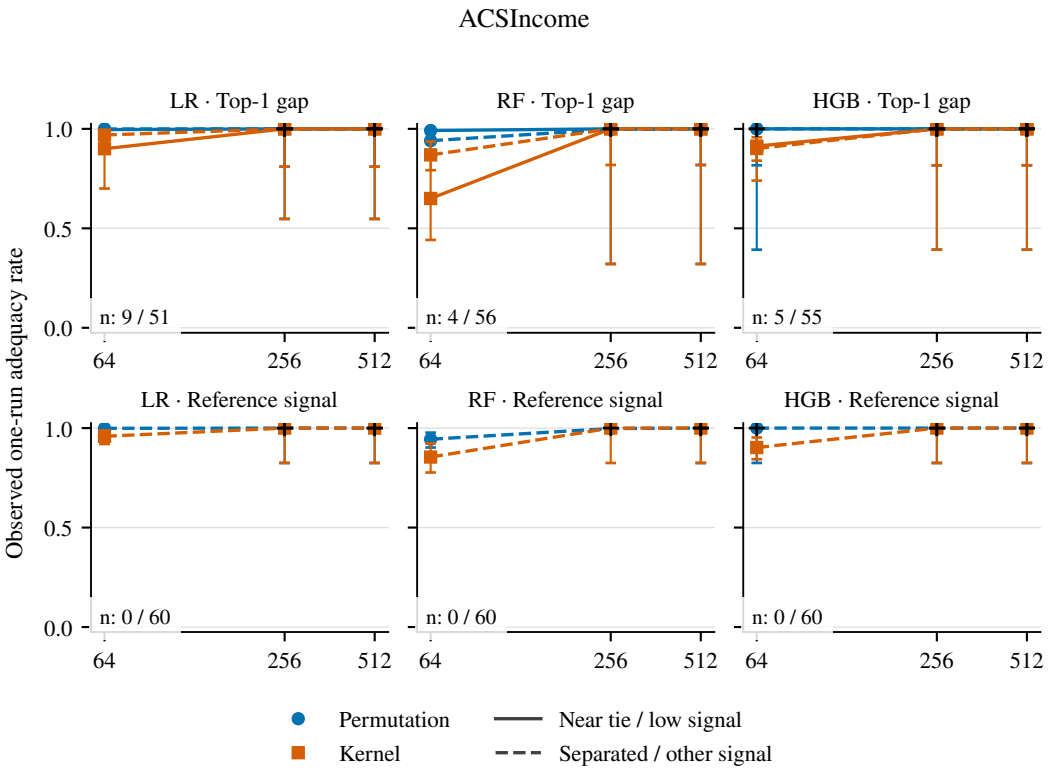


Figure 10: Reference-defined conditions for ACSIncome. Upper row: near-top-1 tie versus separated, using (largest minus second-largest absolute reference attribution)/max(reference L1,0.01) $\leq$0.02. Lower row: low signal (reference L1 <0.01 probability units) versus other signal. Each panel labels independent-input counts in solid/dashed order; counts do not multiply across runs or budgets. Empty strata have no estimate; singleton strata show the observed estimate without an interval. Adequacy requires normalized L1 $\leq$0.20 and signed top-3 overlap $\geq$2/3. These descriptive strata are fixed by the exact reference, not selected using explanation performance. No main input has low signal, so that empty stratum is untested rather than a zero-success finding. Requested work labels do not compare equal actual work. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

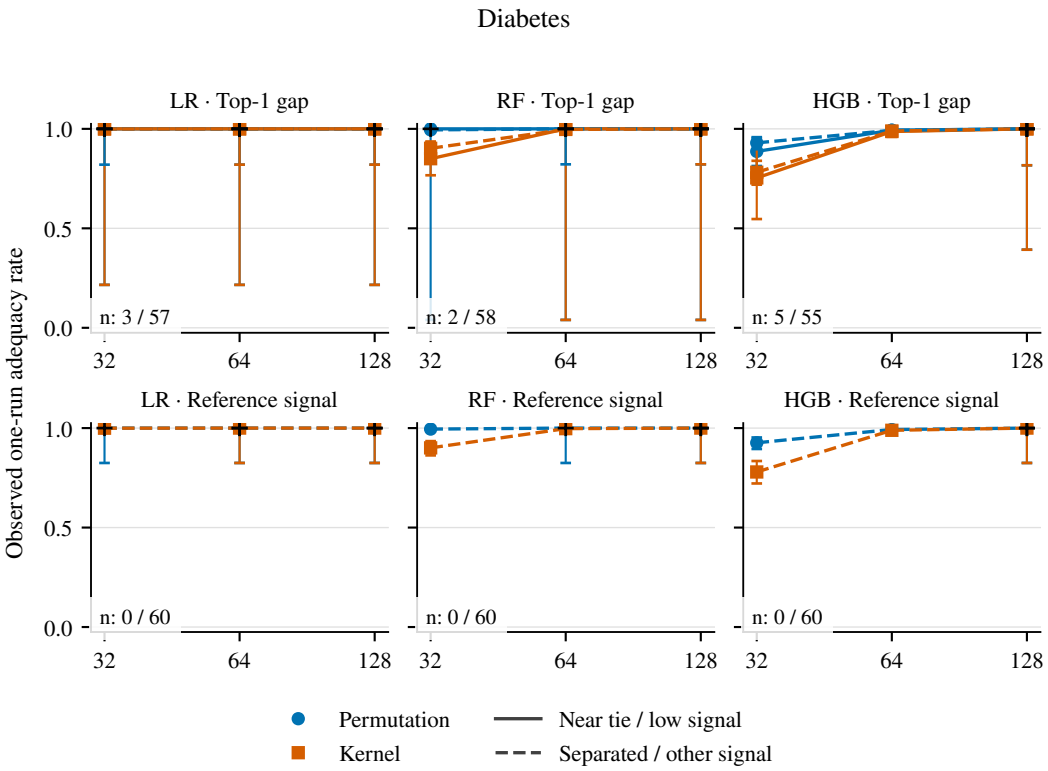


Figure 11: Reference-defined conditions for Diabetes. Upper row: near-top-1 tie versus separated, using (largest minus second-largest absolute reference attribution)/max(reference L1,0.01) $\leq$0.02. Lower row: low signal (reference L1 <0.01 probability units) versus other signal. Each panel labels independent-input counts in solid/dashed order; counts do not multiply across runs or budgets. Empty strata have no estimate; singleton strata show the observed estimate without an interval. Adequacy requires normalized L1 $\leq$0.20 and signed top-3 overlap $\geq$2/3. These descriptive strata are fixed by the exact reference, not selected using explanation performance. No main input has low signal, so that empty stratum is untested rather than a zero-success finding. Requested work labels do not compare equal actual work. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

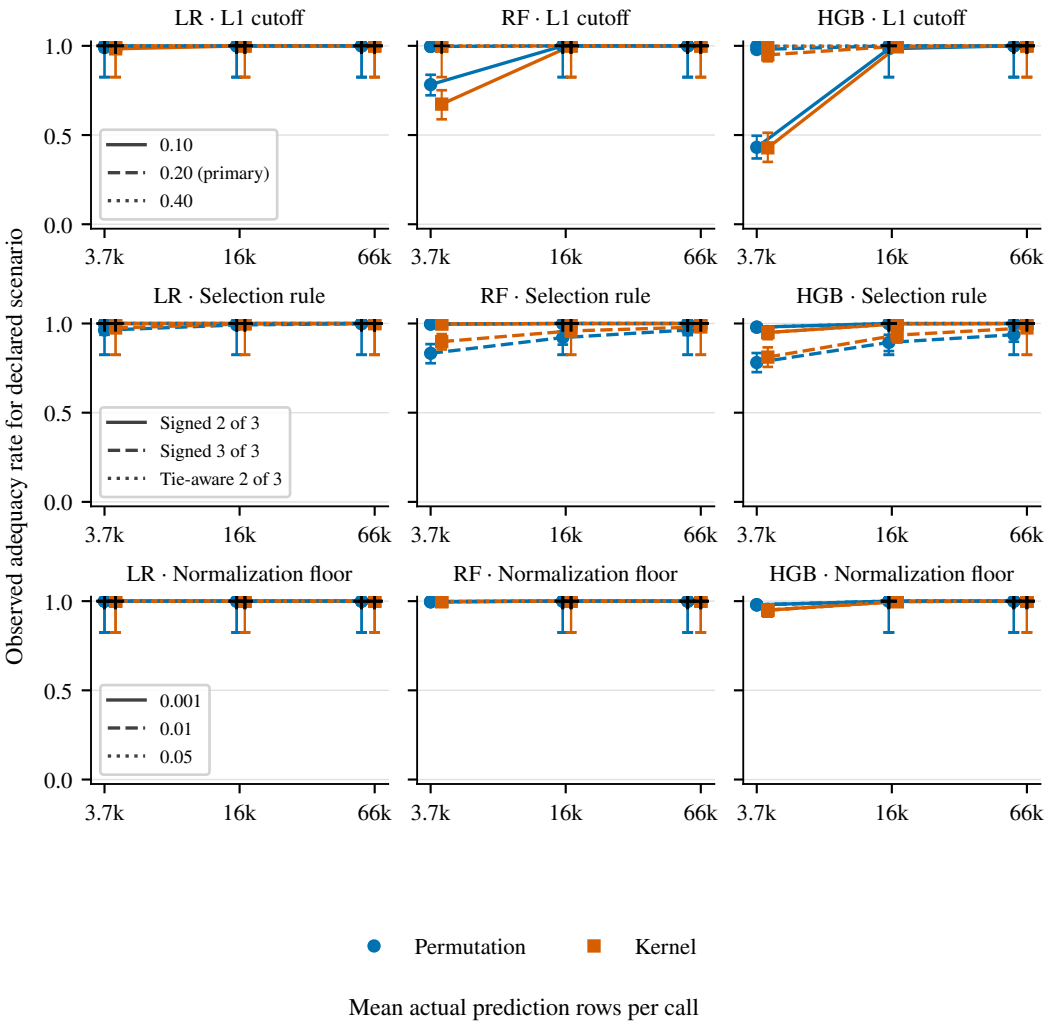


Figure 12: Predeclared adequacy scenarios for German; no scenario is selected after seeing the rates. Top row varies the normalized L1 cutoff (0.10/0.20/0.40), retaining signed top-3 overlap $\geq$2/3 and floor 0.01. Middle row fixes cutoff 0.20/floor 0.01 and compares signed overlap $\geq$2/3 (primary), all three matching signed features, and the analysis-defined 1e-6 probability-unit tie-aware selection rule. Bottom row varies normalization floor (0.001/0.01/0.05 probability units) in max(reference L1,floor), retaining cutoff 0.20 and the same primary signed-top-3 event. These floors change the metric, not the target or explanation calls. All main reference norms exceed 0.05, so these floor curves coincide; weak-signal behavior is untested. Curves use the same runs and are correlated. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

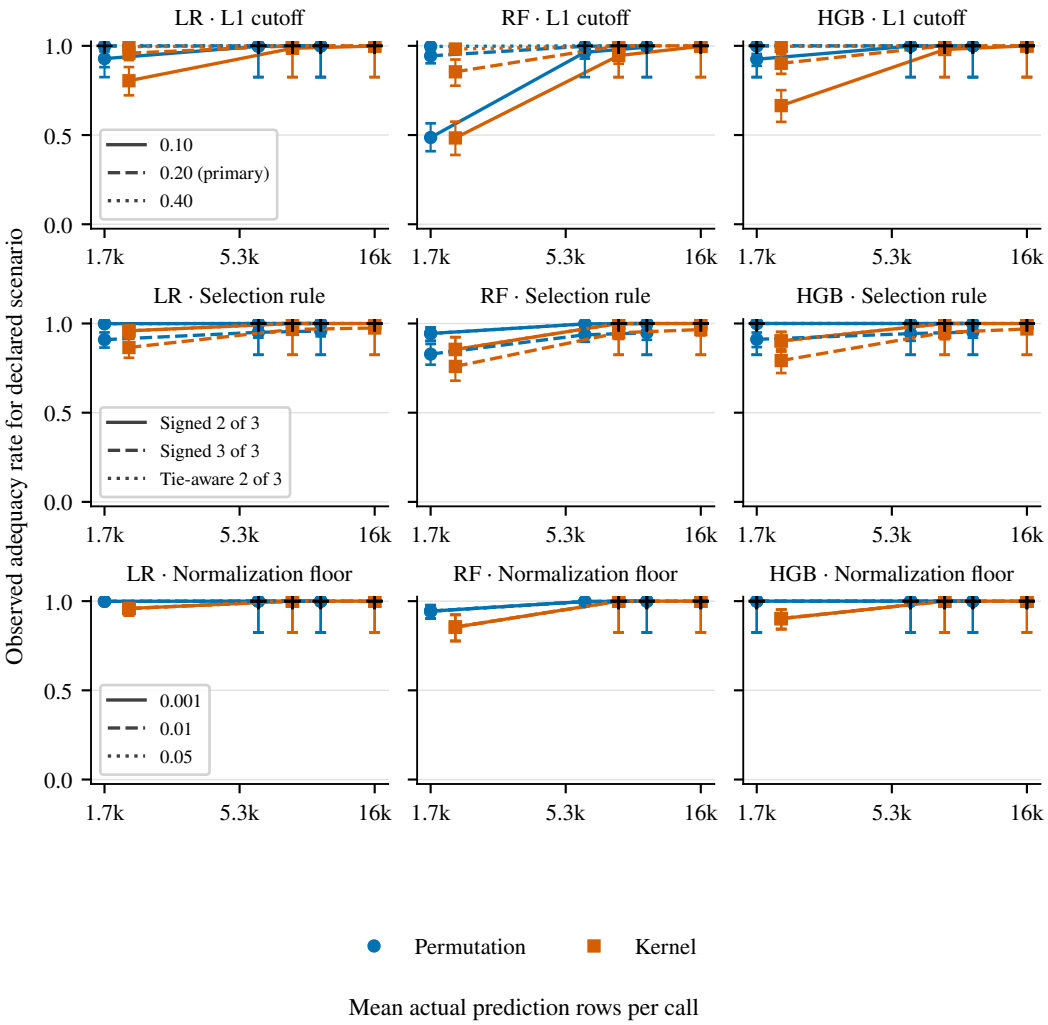


Figure 13: Predeclared adequacy scenarios for ACSIncome; no scenario is selected after seeing the rates. Top row varies the normalized L1 cutoff (0.10/0.20/0.40), retaining signed top-3 overlap $\geq$2/3 and floor 0.01. Middle row fixes cutoff 0.20/floor 0.01 and compares signed overlap $\geq$2/3 (primary), all three matching signed features, and the analysis-defined 1e-6 probability-unit tie-aware selection rule. Bottom row varies normalization floor (0.001/0.01/0.05 probability units) in max(reference L1,floor), retaining cutoff 0.20 and the same primary signed-top-3 event. These floors change the metric, not the target or explanation calls. All main reference norms exceed 0.05, so these floor curves coincide; weak-signal behavior is untested. Curves use the same runs and are correlated. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

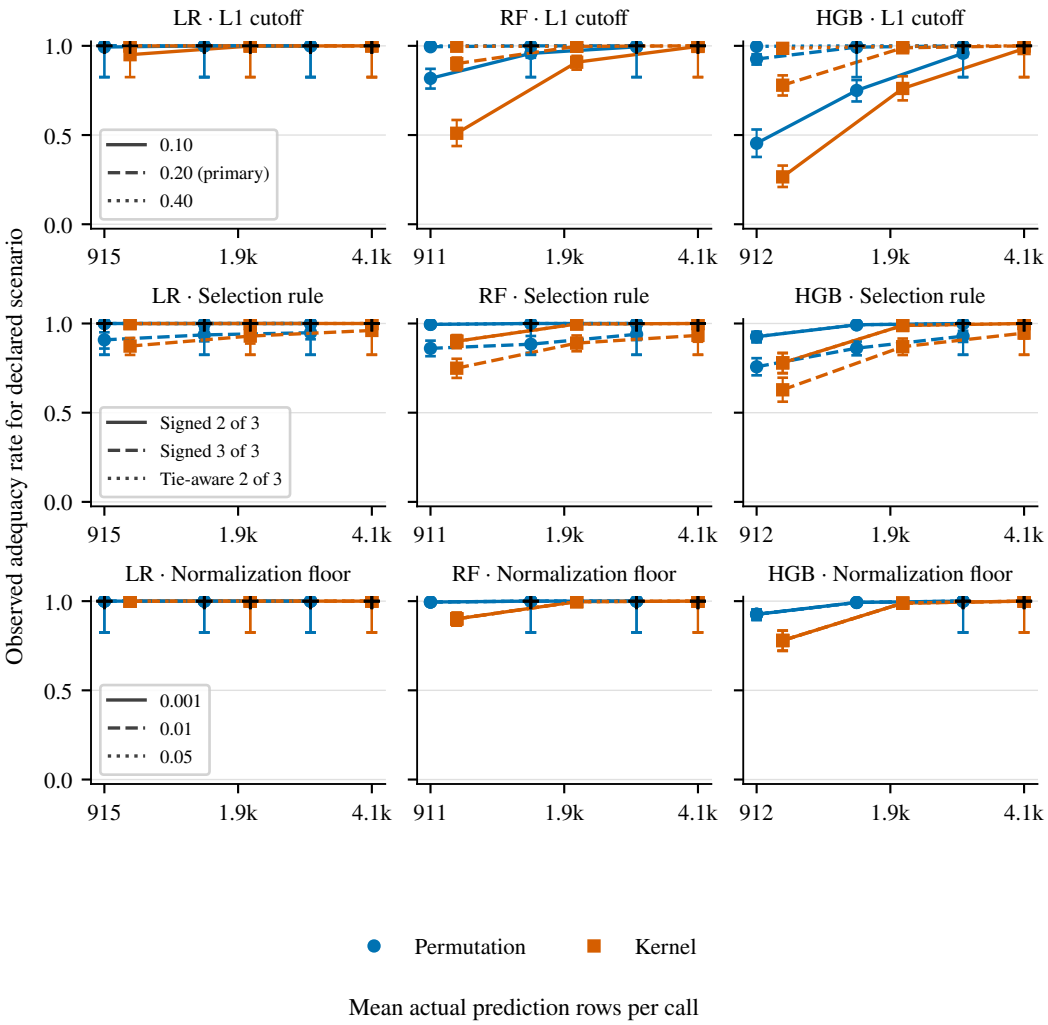


Figure 14: Predeclared adequacy scenarios for Diabetes; no scenario is selected after seeing the rates. Top row varies the normalized L1 cutoff (0.10/0.20/0.40), retaining signed top-3 overlap $\geq$2/3 and floor 0.01. Middle row fixes cutoff 0.20/floor 0.01 and compares signed overlap $\geq$2/3 (primary), all three matching signed features, and the analysis-defined 1e-6 probability-unit tie-aware selection rule. Bottom row varies normalization floor (0.001/0.01/0.05 probability units) in max(reference L1,floor), retaining cutoff 0.20 and the same primary signed-top-3 event. These floors change the metric, not the target or explanation calls. All main reference norms exceed 0.05, so these floor curves coincide; weak-signal behavior is untested. Curves use the same runs and are correlated. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

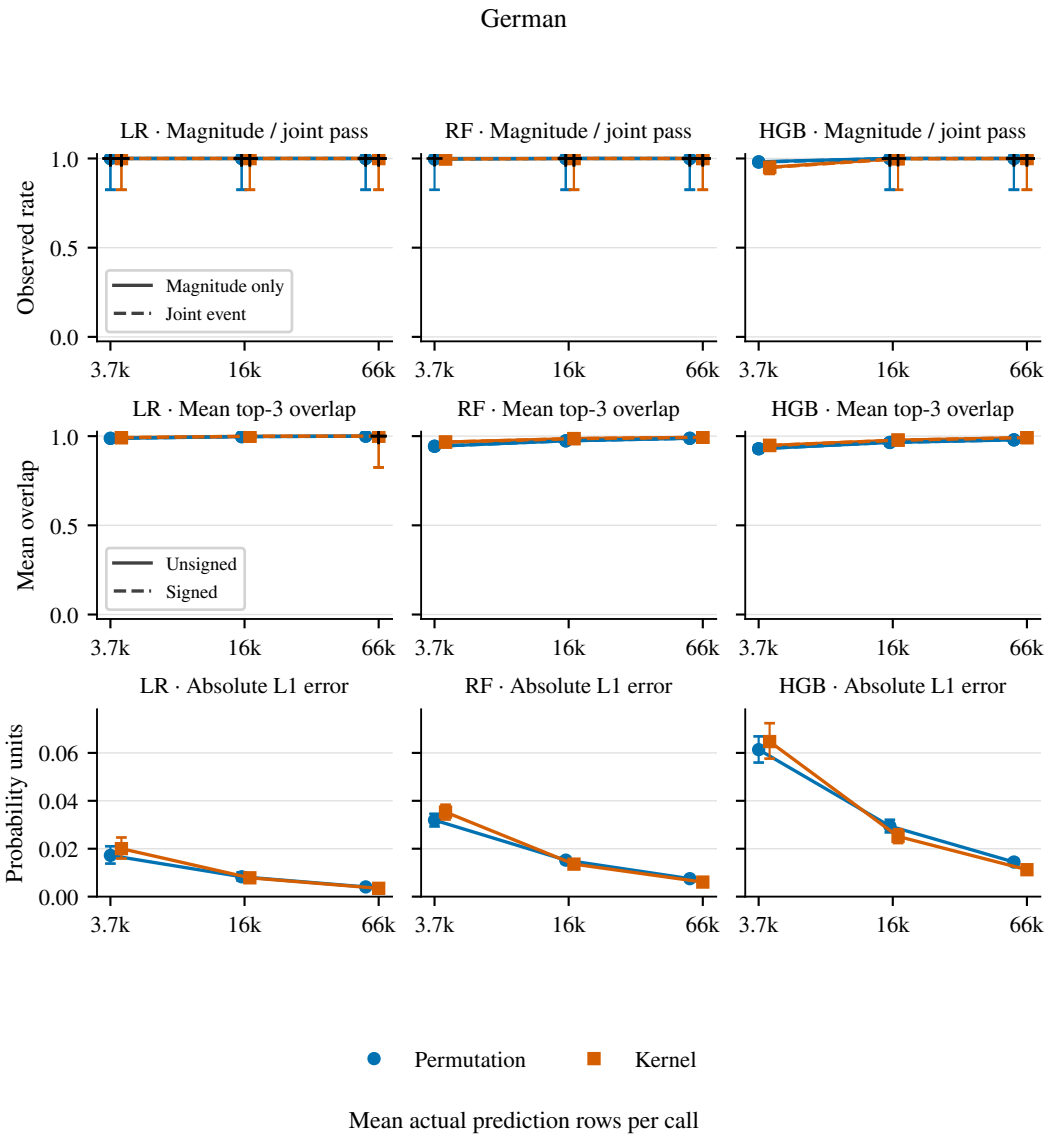


Figure 15: Accuracy components for German. Upper row compares the normalized-magnitude pass probability (L1/max(reference L1,0.01) $\leq$0.20) with the primary joint event, which additionally requires signed top-3 overlap $\geq$2/3. Middle row gives mean unsigned and signed top-3 overlap (matching features or feature-sign tokens, divided by three); these are average overlaps, not pass probabilities. Lower row shows mean absolute L1 error against the exact reference in probability units, without division by the attribution norm. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

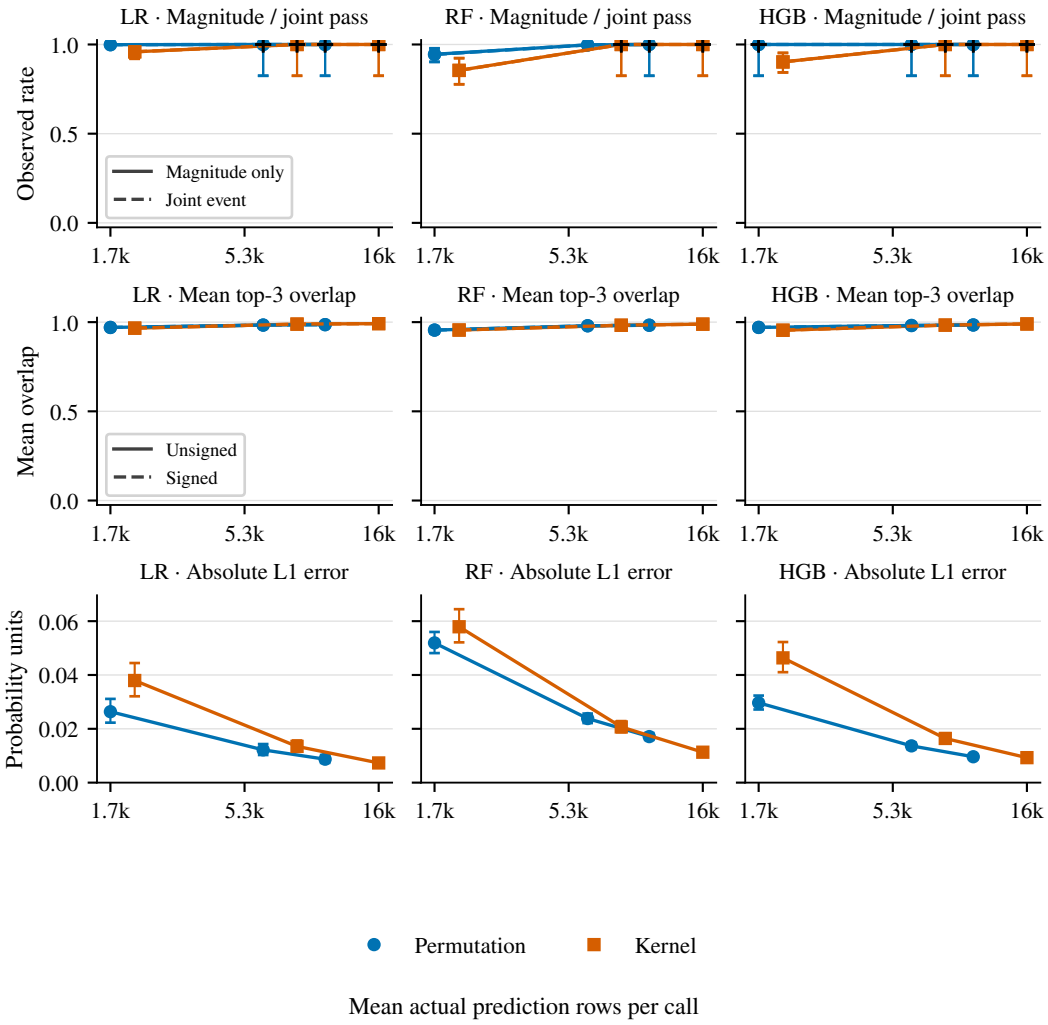


Figure 16: Accuracy components for ACSIncome. Upper row compares the normalized-magnitude pass probability (L1/max(reference L1,0.01) $\leq$0.20) with the primary joint event, which additionally requires signed top-3 overlap $\geq$2/3. Middle row gives mean unsigned and signed top-3 overlap (matching features or feature-sign tokens, divided by three); these are average overlaps, not pass probabilities. Lower row shows mean absolute L1 error against the exact reference in probability units, without division by the attribution norm. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

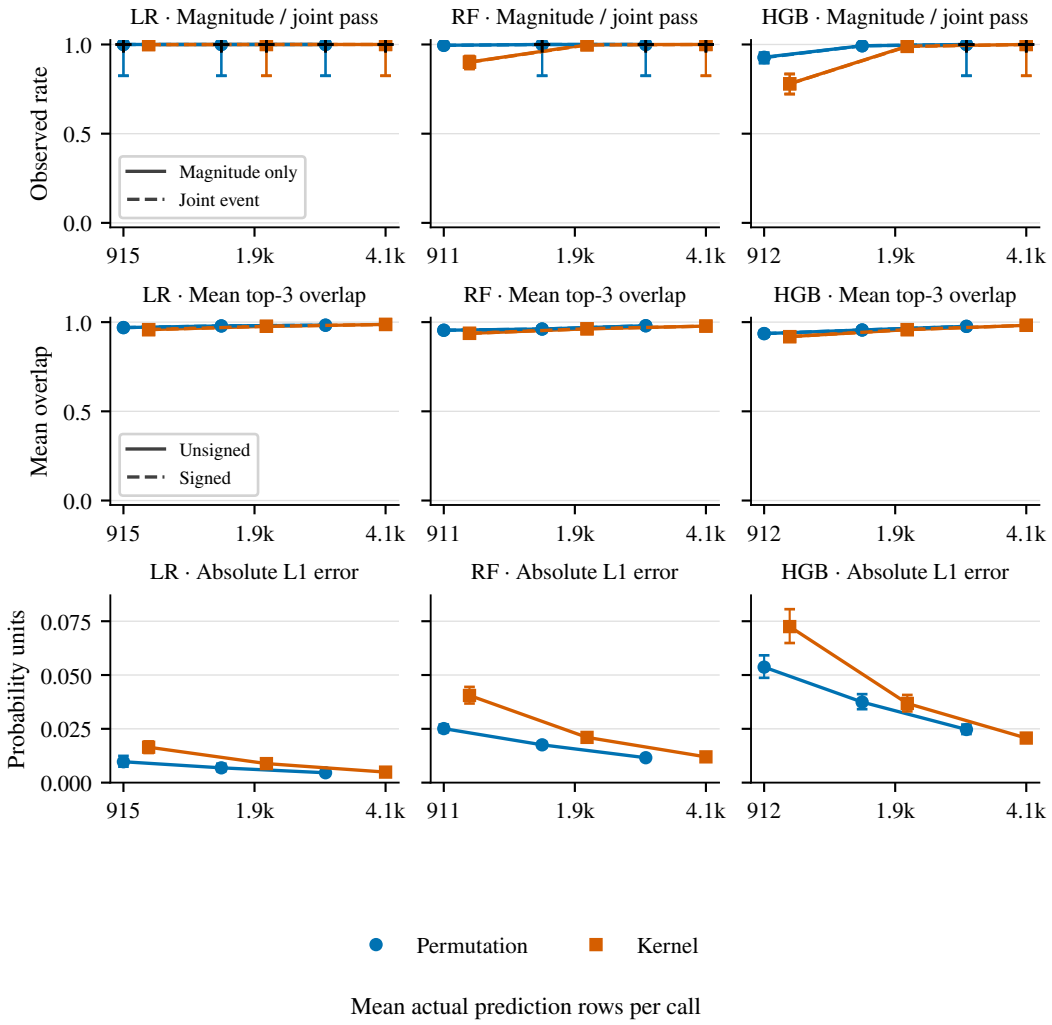


Figure 17: Accuracy components for Diabetes. Upper row compares the normalized-magnitude pass probability (L1/max(reference L1,0.01) $\leq$0.20) with the primary joint event, which additionally requires signed top-3 overlap $\geq$2/3. Middle row gives mean unsigned and signed top-3 overlap (matching features or feature-sign tokens, divided by three); these are average overlaps, not pass probabilities. Lower row shows mean absolute L1 error against the exact reference in probability units, without division by the attribution norm. Each input has 30 independent runs per estimator/budget; the fixed 32-row empirical interventional probability game uses grouped raw features. Bars are analysis-supplied pointwise 95% instance-cluster percentile intervals, conditional on the fitted model, split and background. Black plus markers denote observed all-zero/all-one panels using the supplied two-sided 95% bounded-instance Hoeffding interval instead. Interior intervals are descriptive approximations. No population $\geq$0.95 or individual-input certification follows from the plotted rates.

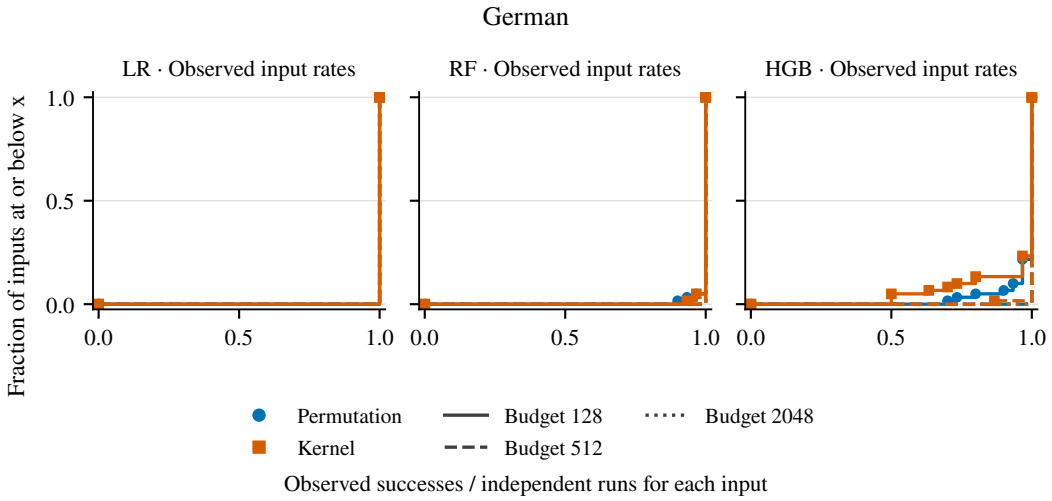


Figure 18: Per-input heterogeneity for German: empirical cumulative distribution of the observed primary-event success fractions among 60 inputs per fitted model, each measured with 30 independent runs per estimator/budget. The event requires L1/max(exact-reference L1,0.01) $\leq$0.20 and signed top-3 overlap $\geq$2/3. Each input contributes once to each curve; vertical masses at zero/one are retained. Curves describe observed finite-run fractions, not the distribution of known true per-input success probabilities. There are no confidence bands or individual-input certifications: even no observed failures in these runs does not establish a 0.95 success probability. Budget labels are requested work, with actual-work/time comparisons supplied separately. The fixed B=32 empirical interventional probability game uses grouped raw features; split/model/background remain fixed.

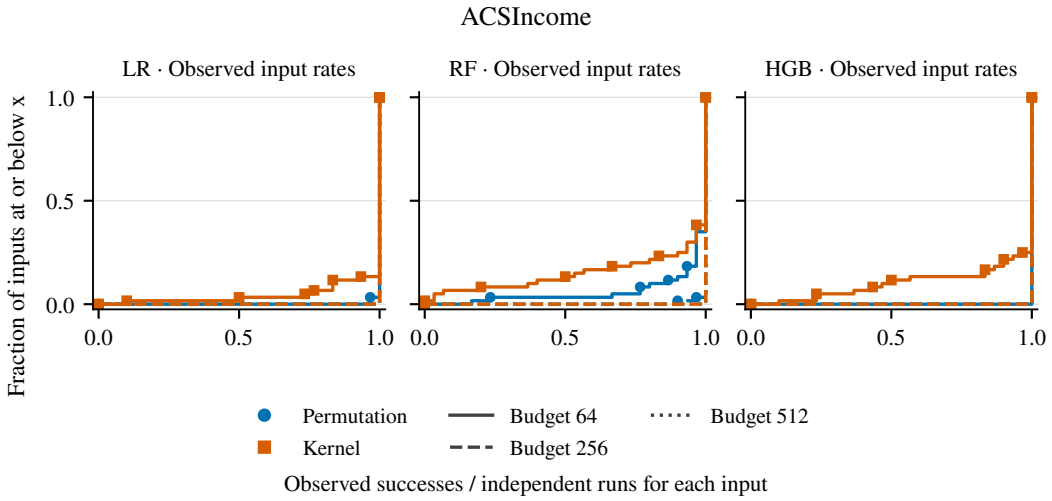


Figure 19: Per-input heterogeneity for ACSIncome: empirical cumulative distribution of the observed primary-event success fractions among 60 inputs per fitted model, each measured with 30 independent runs per estimator/budget. The event requires L1/max(exact-reference L1,0.01) $\leq$0.20 and signed top-3 overlap $\geq$2/3. Each input contributes once to each curve; vertical masses at zero/one are retained. Curves describe observed finite-run fractions, not the distribution of known true per-input success probabilities. There are no confidence bands or individual-input certifications: even no observed failures in these runs does not establish a 0.95 success probability. Budget labels are requested work, with actual-work/time comparisons supplied separately. The fixed B=32 empirical interventional probability game uses grouped raw features; split/model/background remain fixed.

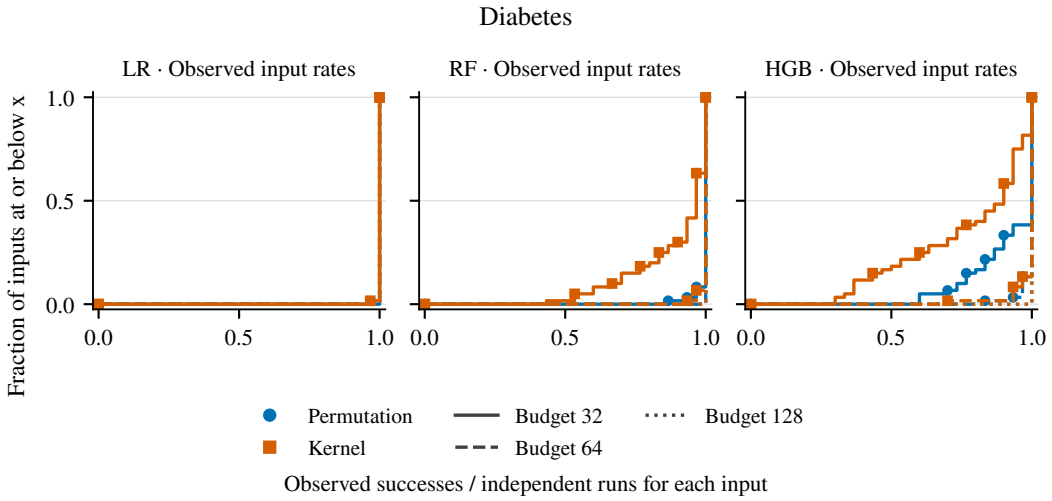


Figure 20: Per-input heterogeneity for Diabetes: empirical cumulative distribution of the observed primary-event success fractions among 60 inputs per fitted model, each measured with 30 independent runs per estimator/budget. The event requires L1/max(exact-reference L1,0.01) $\leq$0.20 and signed top-3 overlap $\geq$2/3. Each input contributes once to each curve; vertical masses at zero/one are retained. Curves describe observed finite-run fractions, not the distribution of known true per-input success probabilities. There are no confidence bands or individual-input certifications: even no observed failures in these runs does not establish a 0.95 success probability. Budget labels are requested work, with actual-work/time comparisons supplied separately. The fixed B=32 empirical interventional probability game uses grouped raw features; split/model/background remain fixed.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Sections 1 and 6 state the measured conditional claims, finite-sample uncertainty and observed failure cases; no deployment or population certificate is asserted.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
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   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Sections 7 and 8 discuss fixed-model/background scope, sparse/empty strata, tolerance choices, survey dependence and interrupted timing.

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